

Structural insights into the recruitment of viral Type 2 IRES to ribosomal preinitiation complex for protein synthesis

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eLife Assessment

This manuscript offers **valuable** structural and mechanistic insights into the structure and assembly of the Type II internal ribosome entry site (IRES) from encephalomyocarditis virus (EMCV) and the translation initiation complex, revealing a direct interaction between the IRES and the 40S ribosomal subunit. While a **solid** cryo-EM method was used, enhancing the overall resolution or adding complementary biochemical data would further improve the clarity and impact of this study. This manuscript will attract researchers in cap-independent translation, host-pathogen interactions, and virology.

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Abstract

Picornaviruses employ internal ribosome entry sites (IRESs) in their genomic RNA to hijack the host's translational machinery. The picornavirus, Encephalomyocarditis virus, employs a type 2 IRES present in its 5'UTR and requires 43S ribosomal preinitiation complex (PIC), the central domain of eukaryotic initiation factor (eIF) 4G, eIF4A, and an essential ITAF (IRES trans-acting factor)-polypyrimidine tract binding protein 1 (PTB1) to form 48S PIC. In this study, we have used cryo-electron microscopy (cryo-EM) to determine the structure of EMCV IRES-bound mammalian 48S PIC in a scanning-arrested closed state at the start codon. The EMCV IRES domains contacts initiator tRNA (tRNA_i) and 40S head at the inter-subunit interface, which reveals an altogether unique mechanism used by viruses to capture host translational machinery for its protein synthesis. The tRNA_i is held away from the 40S body in contrast to canonical cap-dependent translation while the domain I apical region of EMCV IRES mimics 28S rRNA of 60S to interact with 40S ribosomal head proteins-uS13 and uS19. The structural analysis account for numerous biochemical studies on Type 2 IRES and shows how Type 2 IRES interacts with 43S PIC to form 48S PIC. This study provides mechanistic insights for understanding EMCV IRES-mediated translation initiation, which could be extrapolated to other IRESs sharing similar motifs and factor requirements including type 1 viral IRESs.

Introduction

Eukaryotic translation initiation occurs in two different modes-Cap-dependent and Cap-independent. Cap-dependent translation initiation in eukaryotes can be divided into four major steps- (i) formation of 43S ribosomal PIC [40S subunit in complex with eIF1, eIF1A, eIF3, eIF5 and ternary complex (eIF2-GTP-tRNA_i)]; (ii) recruitment of 43S PIC to 5' end of capped mRNA, mediated by eIF4F complex (formed by eIF4E, eIF4G and eIF4A) yielding 48S complex (48S PIC); (iii) scanning of 5' untranslated region (UTR) of mRNA and start codon recognition; and (iv) joining of large 60S ribosomal subunit to form elongation-competent 80S ribosomal complex. Among these, major rate-limiting steps include- the regulation of available tRNA_i as the ternary complex (eIF2.GTP.Met-tRNA_i) and recruitment of mRNA on 43S PIC (Jackson et al 2010 [\[1\]](#); Brito Querido et al 2024 [\[2\]](#)). Cryo-EM studies on eukaryotic canonical 48S PICs from yeast (Hussain et al 2014 [\[3\]](#); Llacer et al 2015 [\[4\]](#)) to humans (Eliseev et al 2018 [\[5\]](#); Simonetti et al 2020 [\[6\]](#); Brito Querido et al 2020 [\[7\]](#); Yi et al 2022 [\[8\]](#); Brito Querido et al 2024 [\[2\]](#); Petrychenko et al 2024 [\[9\]](#)) have revealed various interactions among the initiation factors, mRNA, and ribosome in scanning (P_{OUT} or open) state and scanning-arrested (P_{IN} or closed) states. In mammals, 48S PIC formation is mediated by the interaction of eIF3 and eIF4G (Villa et al 2013 [\[10\]](#)) and recent attempts to understand the mammalian canonical 48S PIC could capture interactions of eIF4 proteins (eIF4G and eIF4A) with eIF3 (Brito Querido et al 2020 [\[7\]](#); Brito Querido et al 2024 [\[2\]](#)).

Alternatively, several positive strand RNA viruses use Internal Ribosome Entry Sites (IRESs), which are internal cis-acting sequences present in 5' UTR of mRNA that drive the assembly of translation initiation complex without the requirement of 7-methylguanosine cap (Martínez-Salas et al 2018 [\[11\]](#)), initially reported in picornaviruses such as poliovirus (PV) and encephalomyocarditis virus (EMCV) RNA genome (Jang et al 1988 [\[12\]](#); Pelletier and Sonenberg, 1988 [\[13\]](#); Trono et al 1988 [\[14\]](#)). As of 2020, the IRESbase database reported 1328 IRESs, out of which 554 were viral IRESs from 198 viruses (Zhao et al 2020 [\[15\]](#)). These RNA elements have unique secondary and tertiary structures, which allows them to hijack the host translational machinery and promote translation initiation internally by recruitment of host ribosome, canonical eIFs, and IRES trans acting factors (ITAFs) via multiple RNA-RNA and RNA-protein interactions (Lee et al 2017 [\[16\]](#); Martínez-Salas et al 2018 [\[11\]](#)). The IRESs lack sequence homology and enfold different structural organization, thus requiring different eIFs for the assembly of 48S PICs (Jackson et al 2010 [\[1\]](#)). The type 1 IRES (represented by PV) and type 2 IRES (EMCV) requires almost all eIFs, except eIF4E or N terminal of eIF4G for the formation of 48S PIC (Lozano and Martinez-Salas, 2015 [\[17\]](#)). However, initiation on the latter does not require scanning for start codon recognition (Kaminski et al 1994 [\[18\]](#); Sweeney et al 2014 [\[19\]](#)). The type 3 IRES (Hepatitis C virus-HCV) does not require eIF4 factors or scanning for 48S PIC formation (Niepmann and Gerresheim 2020 [\[20\]](#)). The type 4 IRES (Cricket paralysis virus-CrPV intergenic IRES) initiates without the requirement of any eIFs and tRNA_i (Johnson et al 2017 [\[21\]](#)). The HCV IRES and CrPV intergenic IRES directly interacts with host 40S ribosome, however they differ in their mechanism of attachment. Where, the CrPV IRES (type 4) mimics anticodon-codon interaction via pseudoknot 1 to bind with ribosome in an elongation-competent state (Petrov et al 2016 [\[22\]](#); Fernandez et al 2014 [\[23\]](#)) the HCV IRES (type 3) binds to the solvent side of 40S subunit by replacing eIF3 and directly interacts with expansion segment (ES7) of 18S rRNA and ribosomal proteins near mRNA exit channel, thus placing the initiation codon at the P-site (Hashem et al 2013 [\[24\]](#); Niepmann and Gerresheim 2020 [\[20\]](#); Brown et al 2022 [\[25\]](#)). While the structures of ribosome-bound type 4 (Spahn et al 2004 [\[26\]](#); Fernandez et al 2014 [\[23\]](#); Muhs et al 2015 [\[27\]](#); Murray et al 2016 [\[28\]](#); Pisareva et al 2018; Acosta-Reyes et al 2019 [\[29\]](#); Abaena et al 2020) and type 3 IRES (Spahn et al 2004 [\[26\]](#); Boehringer et al 2005 [\[30\]](#); Hashem et al 2013 [\[24\]](#); Yamamoto et al 2014 [\[31\]](#); Quade et al 2015 [\[32\]](#); Yamamoto et al 2015 [\[33\]](#); Yokoyama et al 2019 [\[34\]](#); Brown et al 2022 [\[25\]](#)) have been determined, there is no structural information about Type 2 and Type 1 IRES and their mode of recruitment to 48S PIC.

The 5'UTR of EMCV, genera *Cardiovirus* and *Picornaviridae* family ([International Committee on Taxonomy of Viruses Executive Committee, 2020](#)) folds into various stem-loops numbered D-L and, domains H-L, followed by initiation codon at the 834th residue (~450 nucleotides in length) makes a functional IRES moiety ([Carocci et al 2012](#)). It requires the core of 43S PIC, the central domain of eIF4G, eIF4A, and an essential ITAF-polypyrimidine tract binding protein 1 (PTB1) and the presence of eIF4B enhances 48S formation ([Hellen and Wimmer 1995](#); [Martinez-Salaz et al 2018](#); [Sweeney et al 2014](#)). Functional characterization of these domains showed H, and I interact with 40S ribosomal subunit ([Chamond et al 2014](#)), and J-K domain recruits eIF4G, enhanced by eIF4A ([Pestova et al 1996](#); [Lomakin et al 2000](#)). The structure of J-K domain-eIF4G1(HEAT1)-eIF4A revealed that positively charged patches on the eIF4G1-HEAT1 domain interact with two separated negatively charged clefts (in J and St domain) without perturbing its innate function of recruiting eIF4A ([Imai et al 2016](#); [Imai et al 2023](#)). The binding sites for PTB1 were revealed to be dispersed, encompassing H to L domains ([Kafasla et al 2009](#)). These domains function as a single entity to form 48S PIC and any mutation in conserved IRES motifs can drastically affect the translation rates ([Fernández-Miragall et al 2009](#); [Fernández et al 2011](#)). For example, the RAAA and GNRA loop in the I domain of type 2 IRES are crucial for IRES activity ([Robertson et al 1999](#)), and the GNRA loop is also found in other IRES families such as type 1 (poliovirus) and type 5 (Aichi virus A) picornaviruses ([Abdullah et al 2023](#)). The inherent flexibility within the IRES domains provide a challenge for structural studies of full length IRES and to capture IRES in the context of 48S PIC. EMCV IRES can independently interact with 40S subunit without any eIFs ([Chamond et al 2014](#)) and does not require scanning to recognize the start codon ([Pestova et al 1996](#)), unlike type 1 IRESs which scans for the start codon ([Sweeney et al 2014](#)). Besides, EMCV IRES can form 48S PIC with HEAT1-eIF4G without the requirement of eIF4G residues 1015-1104 that are known to interact with eIF3 ([Lomakin et al 2000](#)). These residues are indispensable in case of canonical initiation on capped mRNAs and for type 1 IRESs ([Villa et al 2013](#); [Sweeney et al 2014](#)). However, how EMCV IRES interacts with 40S subunit or what are the molecular interactions essential to form the EMCV IRES-48S PIC remains a question.

In this study we have used pull down assay to isolate EMCV IRES-bound 48S PIC from Rabbit Reticulocyte Lysate (RRL) and subjected the complex to cryo-electron microscopy (Cryo-EM) analysis. The cryo-EM map of EMCV IRES-bound 48S PIC, henceforth mentioned as EMCV IRES-48S PIC, shows densities corresponding to EMCV IRES domains that reveals how EMCV IRES contacts the ternary complex and 40S head at the inter-subunit interface. The structural details presented here accounts for numerous biochemical studies that have been reported earlier on Type 2 IRES. Furthermore, the structural analysis suggests how Type 2 IRES would interact with 43S PIC to form 48S PIC. Importantly, the study reveals a unique strategy used by viral IRES to capture host translational apparatus for making viral polypeptide.

Results

1. Overview and features of EMCV IRES-48S PIC

To isolate 48S PIC on EMCV IRES, we used a Talon affinity based pull down from nuclease-treated RRL. EMCV IRES RNA used harboured residues from 280 to 905 with AUG (start codon) at 834th position. PTB1 was recombinantly overexpressed with a N-terminal 6X His tag, followed by a 3C protease cleavage site. PTB1 was incubated with the IRES, followed by RRL addition and 48S PIC was stalled using GMP-PnP. The complex was eluted from Talon matrix employing 3C protease cleavage and pelleted (Sup. Fig. 1.1 A). The EMCV IRES-48S PIC was subsequently analysed by Cryo-EM. The processed data yielded 3 major classes- (i) 40S without any factors-Map A, (ii) 40S-IRES-tRNA_f-Map B, and (iii) 40S-IRES-ternary complex-Map B1, namely, EMCV IRES-48S PIC (Sup. Fig. 1.1 B). Map B and Map B1 has an overall resolution of 4.6Å and 5.0Å, respectively (Sup. Fig. 1.1 B). The

core of 40S is at around 4.0Å and the local resolution across the maps were largely in the range of 4.0-8Å (Sup. Figs. 1.2 A and B). Only the extreme tip of beak of 40S in Map B1 (Sup. Fig. 1.2 A) and ends of IRES and eIF2γ is around 12Å in Map B2 (Sup. Figs 1.2 B).

The cryo-EM densities in EMCV IRES-48S PIC corresponds to 40S, tRNA_i, eIF2α, eIF2γ and RNA in the mRNA channel along with an extra density connecting 40S ribosomal head to tRNA_i (**Fig. 1A-B**). However, it lacks distinct density for PTB1, eIF4G, eIF4A, and eIF3 and hence these factors are not modelled. Since nuclease-treated RRL lacked endogenous RNA, the presence of density for mRNA in the channel indicates trapping of EMCV IRES in 48S PIC. Also, the density connecting 40S head to tRNA_i in Map B and B1 could be assigned to a double stranded RNA structure found in EMCV IRES (**Fig. 1A-B**).

Besides, an extra density is evident at the mRNA entry site, contacting 40S ribosomal proteins-uS3, eS10, and uS5 and h16 (Helix 16) of 18S rRNA (Sup. Fig. 1.2 C). In the human 48S PIC, this region positions eIF3g-RRM (RNA recognition motif) and downstream ORF (Sup. Fig.1.2 D) (Brito Querido et al 2020; Brito Querido et al 2024). We anticipate this extra density in EMCV IRES-48S PIC could be contributed by downstream ORF, however, it's difficult to assign it to eIF3g-RRM due to lack of eIF3 (core/peripheral) density. Furthermore, our recent study on interaction of yeast eIF4B with 40S ribosome suggested occupancy of eIF4B (N terminal-RRM) in the same region (Datey et al 2025), which opens up the possibility of mammalian eIF4B-NT-RRM to bind to this region in the absence of eIF3, although in the human 48S PIC eIF4B was tentatively positioned slightly away from this location (Brito Querido et al 2024). An interesting possibility could be positioning one of the RRM domains of PTB1 bound to UCUUU sequence of 18S rRNA (PTB1 can bind to UCUUU sequence- Maris et al 2020) present at the tip of h16 (528-532 nucleotides) of 18S rRNA. An extra density in the same position is also present in Map A (40S without EMCV IRES or factors-Sup. Fig. 1.2 E) as well, and PTB1 was used as a bait for the pull-down, therefore, this density may be contributed by PTB1-RRM interacting with h16 of 18S rRNA.

2. The EMCV IRES-48S PIC is trapped in a closed conformation

In EMCV IRES, A-834-G-836 forms the start codon (Kaminski et al 1994; Hellen and Wimmer 1995; Pestova et al 1996) and it is present in a Kozak context (CRCCAUGG; R is a purine) (Kozak 1989), where the -3 position is A-831 and +4 is G-837 (Sup. Fig. 2D). Based on these, we anticipate the 834th AUG base-paired to the anticodon (**Fig. 2A**) and the flanking residues were added as per the sequence (Sup. Fig. 2D). The recognition of the start codon at the P-site by tRNA_i leads to accommodation of ternary complex and 48S PIC adopts a P_{IN} state in contrast to P_{OUT} state observed during scanning (Llacer et al 2015; Llacer et al 2021; Yi et al 2022; Petrychenko et al 2024). A distinct difference of 7.0Å could be seen by comparing the tRNA_i position of EMCV IRES-48S PIC with that in the open state (**Fig. 2B**), depicting a P_{IN} state of tRNA_i. The entrapment of start codon in the P-site also evokes closure of the mRNA latch formed by 18S rRNA helices-h34 of 40S head and h18 of body (Hussain et al 2014; Llacer et al 2015; Hinnebusch 2017). The h34 of 18S rRNA of EMCV IRES-48S PIC is shifted by 9Å as compared to the human canonical open state 48S PIC (PDB-7QP6, Yi et al 2022). This conformation of 18S rRNA correlates well with that of closed 48S PIC (PDB-7QP7, Yi et al 2022), suggesting EMCV IRES-48S PIC was captured in a closed state (**Fig. 1C**). This closed conformation locks the mRNA in the channel formed within 40S head and body. On the other hand, the conformation of 18S rRNA in Map A (40S ribosome without initiation factors) shows an open state (Sup. Fig. 2E). The transition from open (PDB Id-7QP6) to closed states (PDB Id-7QP7) structures is also accompanied by an upward shift of N terminal domain of eS17 (connecting the head to body) by 10Å (Yi et al 2022). The EMCV IRES-48S PIC structure correlates with the conformation of eS17 in the human 48S-closed PIC, that is eS17 N-terminal helix associated with the ribosomal head shifts upward by 10Å, keeping C-terminal domain position constant (**Fig. 2D**; Sup. Fig. 2F).

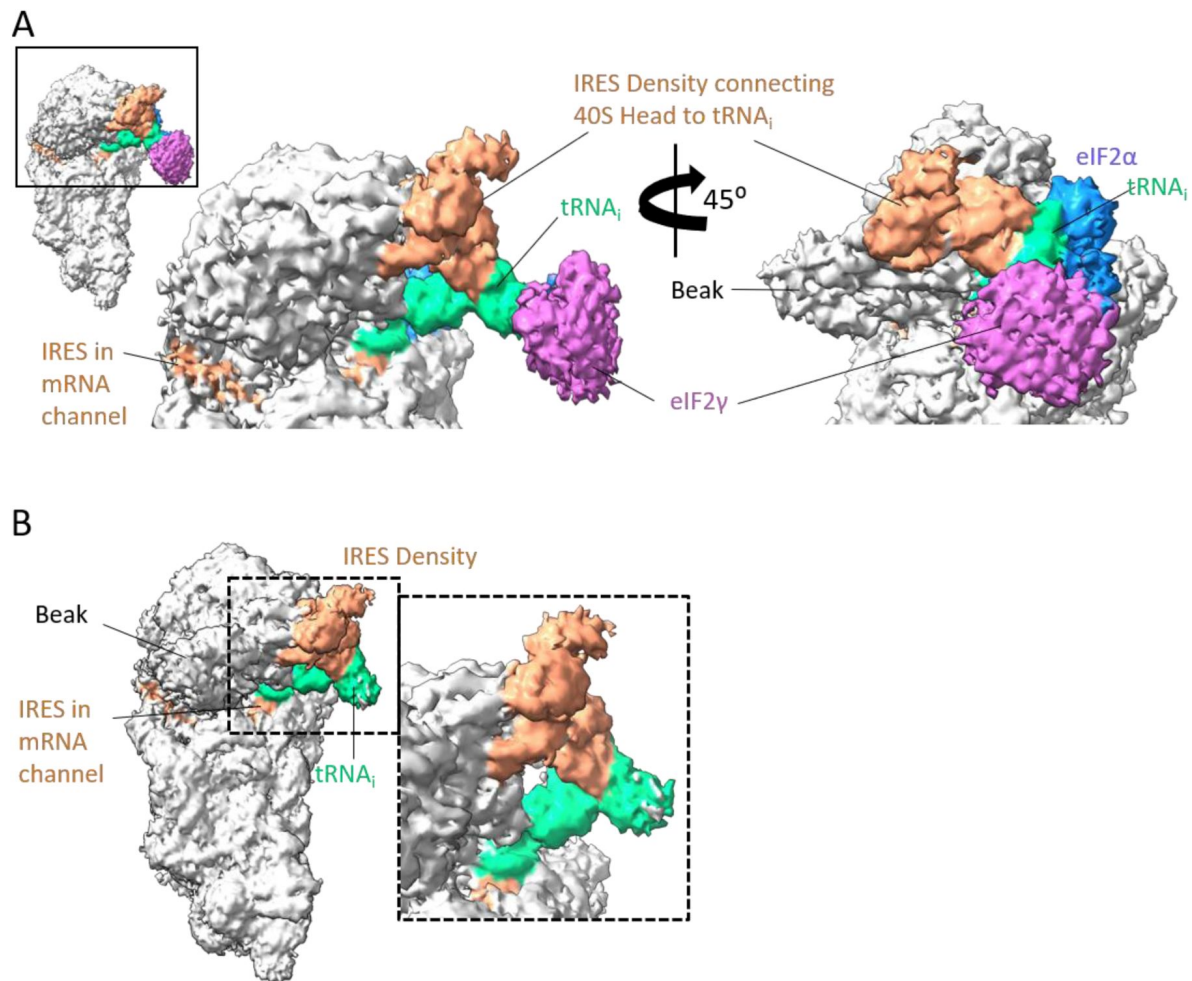


Figure 1

(A) Different views of EMCV IRES-48S PIC (Map B1) by 45° rotation along one axis. The map shows densities assigned to 40S ribosome, RNA (in channel), ternary complex and IRES density at the inter-subunit region of head. **(B)** View of Map B and zoomed view of densities corresponding to IRES and tRNA_i.

Data collection	Map A- 40S without factors	Map B- 40S- tRNA _i -EMCV IRES	Map B1- 40S- tRNA _i -EMCV IRES-eIF2 α - eIF2 γ
Microscope	Talos Arctica Gatan K2 Summit Direct Detector 36000X 200 50-55 -2 to -0.5 1.17 22549		
Camera			
Magnification			
Voltage (kV)			
Electron dose (e ⁻ /Å ²)			
Defocus range			
Pixel size (Å)			
Number of micrographs			
Cryo-EM reconstruction			
Final number of particles	125503	55231	28439
Point group symmetry	C1	C1	C1
Fsc threshold	0.143	0.143	0.143
Map overall resolution(Å)	4.51	4.55	5.01
Resolution metric	Gold-Standard FSC	Gold-Standard FSC	Gold-Standard FSC
Sharpening B factor(Å ²)	-153.5	-117.9	-122.3
Atomic model refinement			
Resolution (0.5) (Å)	5.0	5.4	6.0
D FSC model (0/0.143/0.5)	4.4/4.5/5.0	4.4/4.5/5.4	4.9/4.9/6.0
Initial models used	6YAN	6YAN; 8OZO	6YAN; 8OZO
CC overall			
CC(mask)/(box)/(peaks)/(volume)	0.73/0.83/0.72/ 0.66	0.77/0.84/0.71 /0.77	0.76/0.84/0.7 0/0.75
Molprobrity score	2.3	2.28	2.31
Clash score	19.08	19.09	21.24
No. of atoms/ No. of residues			
Chains	39	41	43
Total atoms (Hydrogens:0)	76455	80057	85830
No. of residues- Proteins/Nucleotides	4916/1744	4918/1912	5660/1912
Bond (RMSD) Lengths (Å)	0.002	0.003	0.003
Bond (RMSD) Angles (°)	0.658	0.650	0.665
Ramachandran plot (%)			
Outliers	0.21	0.27	0.21
Allowed	8.94	8.21	7.90
Favored	90.86	91.52	91.89
Rotamer outliers (%)	0.02	0.02	0.00
C β outliers (%)	0.00	0.00	0.00
CaBLAM outliers (%)	6.09	5.82	5.46

Table 1

Cryo-EM data and model statistics

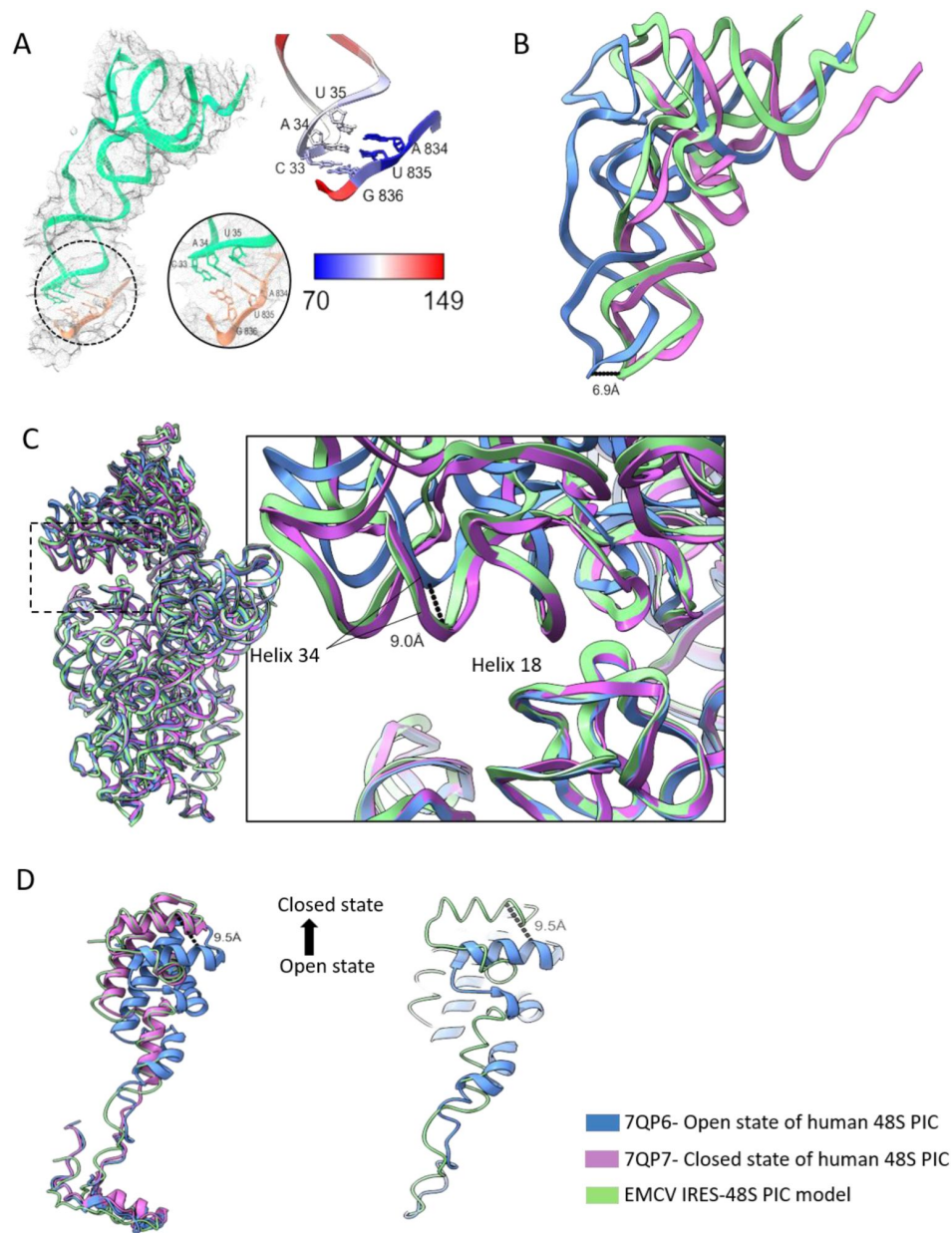


Figure 2

(A) Fitting of tRNAi-base paired to start codon AUG (left). Zoomed view of codon-anticodon interaction (middle), and B-factor for codon-anticodon interaction (right). **(B)** The tRNAi in EMCV IRES-48S PIC is in PIN state similar to that in human 48S PIC PIN state (PDB Id-7QP7) and its anticodon shows ~7 Å shift as compared to human 48S PIC open state (PDB Id-7QP6). **(C)** Comparison of ribosomal conformation (18S rRNA) of EMCV IRES 48S PIC with Human open and closed state (PDB Id-7QP6 and 7QP7, respectively). Focusing on the entry site and the helices governing latch conformation-Helix 34 moves toward helix 18 by 9 Å. **(D)** Movement of eS17 in open and closed states of ribosome. Zoomed view of comparison between eS17 in EMCV IRES 48S PIC and Human 48S PIC open state-showing an upward shift of the N terminal domain by 10 Å.

3. The IRES density connecting 40S head to tRNA_i is contributed by the I domain apical part

In EMCV IRES-48S PIC complex we observed a density, likely for double stranded RNA, connecting 40S head to the tRNA_i elbow region (**Fig. 3A**). The extra density interacts with the elbow region and acceptor stem of tRNA_i and ribosomal proteins-uS19 (RPS15) and uS13 (RPS18) (*Nomenclature-Ban et al 2014*). The EMCV IRES contains 4 major stem-loops (H-L) in the functional IRES region (**Fig. 3B**) (Hellen and Wimmer 1995). Among these, domain H and I have been shown to interact with 40S (*Chamond et al 2014*) and mutations of conserved residues in these domains severely compromised translation on EMCV IRES (López de Quinto and Martínez-Salas 1997; Hellen and Wimmer 1995). We reasoned that H and I domain may contribute to the double stranded RNA density emanating from 40S head. The density architecture (**Fig. 3C**) could be interpreted as a long main stem (S1) extending away from 40S where the base is anchored to the ribosome by two branches (B1 and B2), and to tRNA_i by one branch (B3), further divided into two sub-branches (B3a and B3b). Visualizing the IRES RNAfold determined-secondary structure (**Fig. 3D**)-which correlates with the experimental structure proposed in *Duke et al 1992*, this architecture could be contributed by the apical part of I domain.

To determine the tertiary structure of the I domain apex, the IRES region from nucleotide 507-619 was modelled using AlphaFold3 (Sup. Fig. 3A). The model was decomposed and reconstructed based on the best-fit in the obtained density for the IRES using Chimera and Coot (Sup. Fig. 3A). The final model was generated after multiple rounds of geometry correction and real space refinements (**Fig. 3E**). The final model holds a correlation coefficient of 0.8 with respect to the map (Sup. Fig. 3B), where B1 is AAG loop, B2 is CAAA loop, B3a-GCGA loop, B3b-C rich loop1, and S1 is the major double stranded stem of I domain. The B-factor of the modelled IRES largely ranges from 124 to 200 (Sup. Fig. 3C). The AlphaFold3 predicted model of H domain could not fit properly in the density (Sup. Fig. 3D). The J-K domain adopts a Y-shaped structure and its placement of its Cryo-EM (PDB Id-8HUJ) or NMR (PDB Id-2NBX) structure in the density would clash with 40S (Sup. Fig. 3E). Moreover, in context to EMCV IRES-48S PIC, J-K domain binds eIF4G and the location of eIF4G has been mapped close to ES6 of 18S rRNA, located near the left foot 40S ribosome (*Yu et al 2011*). The I domain apex model in EMCV IRES-48S PIC shows the RAAA and AAG motif contacts uS19 and uS13, and the GNRA loop with tRNA_i (**Fig. 4A**). These interactions with 40S head and tRNA_i could be facilitated by the long length and flexible nature of I domain.

4. The interaction of I domain of EMCV IRES with ribosomal proteins and initiator tRNA

The I domain is the longest domain in EMCV IRES, which harbours important motifs such as GNRA, RAAA and C rich, crucial for IRES activity (*Roberts and Belsham, 1997*; *Fernández-Miragall and Martínez-Salas 2003*) and conserved across all cardioviruses (*Hellen and Wimmer 1995*). Among these, the CAAA and AAG motifs shares potential contact sites with uS13 and uS19. The alpha helix (100-117 residues) of uS13 contacts the IRES element at the B3 stem (connecting the GNRA loop to RAAA loop) (**Fig. 4B**). uS19 contacts the IRES majorly at its CAAA motif through multiple sites involving N-terminal residues, residues-67 to 75, C-terminal-102 to 124 (**Fig. 4C**). These regions of uS13 and uS19 is rich in basic residues, which might interact with the negatively charged backbone of the IRES element (**Fig. 4B-C**). The role of uS13 and uS19 also involves formation of inter-subunit bridges during 60S joining to form elongation competent 80S complexes. uS13 interacts with uL5 (RPL11) and uS19 with helix 38 or h38 (1748-1778, in humans) in 28S rRNA to form inter-subunit bridges-B1b/c and B1a, respectively (Sup. Fig. 4A) (*Ben-Shem et*

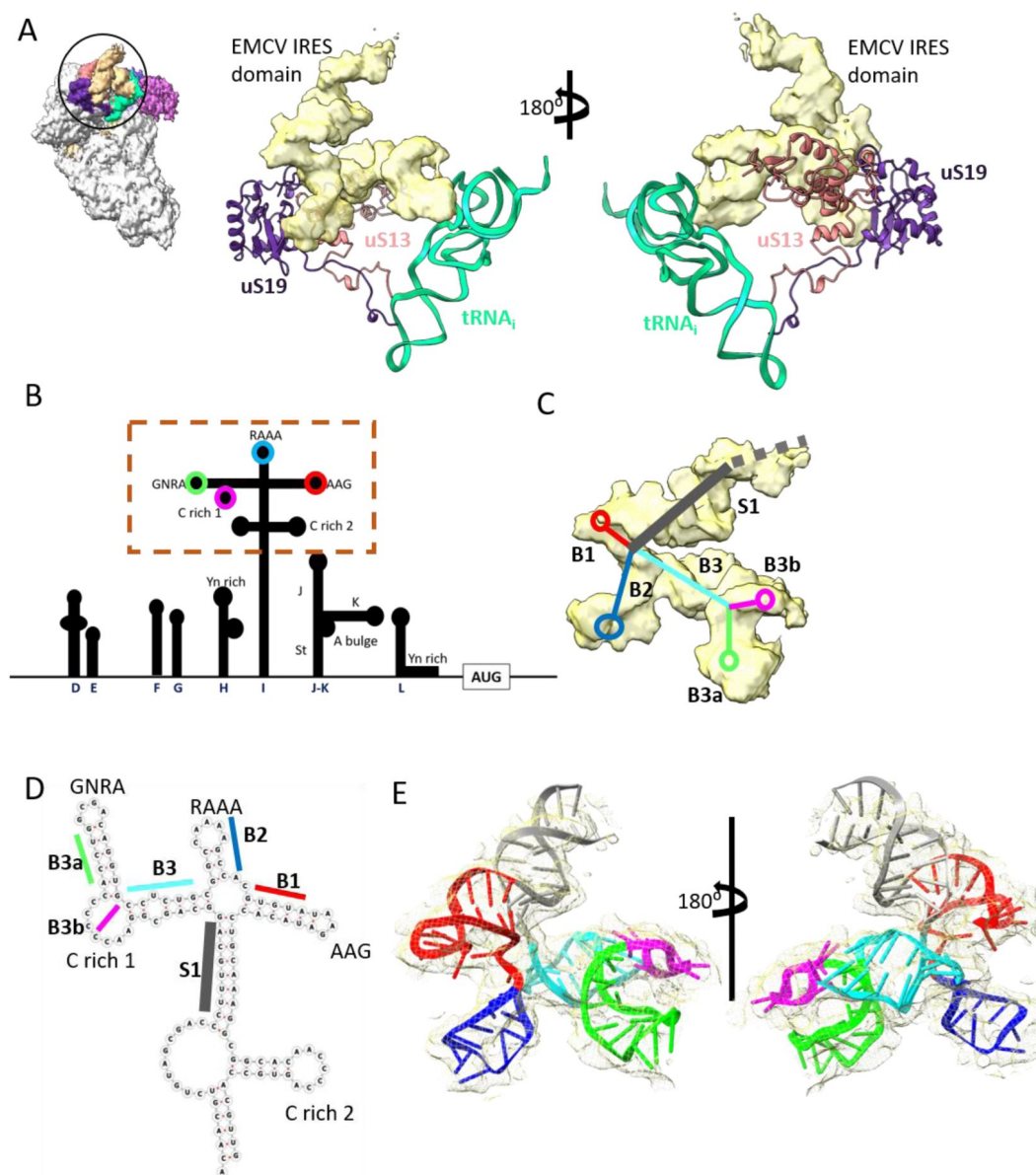


Figure 3

(A) Extra density connecting the head of 40S to tRNA_i elbow region is contributed by EMCV IRES RNA in Map B1. Rotated views of RNA density showing its connection to the 40S head via uS13 and uS19 and to tRNA_i via its elbow region. (B) Organization of EMCV IRES domains from D-L, where H-L makes the functional IRES moiety (Hellen and Wimmer et al 1995). (C) Deciphering the architecture of the obtained IRES density. The density could be interpreted as a long main stem (S1) extending away from the ribosome where the base is anchored to the ribosome by two branches (B1 and B2), and to tRNA_i by one branch (B3), further divided into two sub-branches (B3a and B3b). (D) Secondary structure of apical region of domain I (made using RNAfold) marking the Stem and branches, along with imported loops. (E) Fitting of I domain apex in the density (Rotated views). The sub-domains are coloured as proposed in Fig 3C-D.

al 2011 [↗](#); Bowen et al 2015 [↗](#); Khatter et al 2015 [↗](#)). These interactions are dynamic owing to ribosomal subunit rotation and swivelling during 80S ribosomal translocation states (Khatter et al 2015 [↗](#)).

On superimposition of 80S ribosomal structure to EMCV IRES-48S PIC model, the IRES density clashes with the position of uL5 and h38 of 28S rRNA (Sup. Fig. 4B), suggesting repositioning of IRES domain from 40S head during 48S to 80S transition. Interestingly, the h38 residues interacting with uS19 shares considerable similarity in sequence to I domain apex in the EMCV IRES (Fig. 4D [↗](#)), suggesting the I domain apex of EMCV IRES could mimic h38 (60S)-40S interaction (Fig. 4D [↗](#)). The similarity of h38 with the I domain residues provides additional support for annotation of I domain apex in the density.

The GCGA (GNRA motif-where N is any nucleotide and R is a purine) is known for long range RNA-RNA interactions, widespread in ribosomal RNA and in some catalytic RNAs. It forms a characteristic U-turn structure (Fiore and Nesbitt 2013 [↗](#)) and interacts with minor grooves of helical RNA elements (Sup. Fig. 4C-Reiter et al 2010 [↗](#)). A single point mutation within this tetraloop (GCGA to GCGC) severely reduced the IRES activity, suggesting its essential for IRES activity (Roberts and Belsham, 1997 [↗](#); Fernández-Miragall and Martínez-Salas 2003 [↗](#)). The density extending from the elbow region of tRNA_i could fit in the characteristic U-turn, adopted by conventional GNRA motifs. In EMCV IRES, the GNRA motif is represented by GCGA loop, preceded by a C rich region and in close contact with the tRNA_i elbow and acceptor stem (Fig. 4E [↗](#)). Thus, we infer that EMCV IRES interacts with tRNA_i by virtue of its GCGA loop.

5. The position of eIF2-ternary complex is shifted towards 40S head in EMCV IRES-48S PIC in contrast to canonical 48S PIC

We could fit eIF2 α and eIF2 γ in their respective densities in Map B1 (Fig. 5A [↗](#)). Focussed classification or refinement did not yield any distinct density corresponding to the position of eIF2 β , probably due to the flexibility associated with repositioning of eIF2 β during transition from open to closed complexes (Llacer et al 2015 [↗](#); Llacer et al 2021 [↗](#); Yi et al 2022 [↗](#); Petrychenko et al 2024 [↗](#)). Previous reports on EMCV IRES suggested its direct interaction with eIF2 (Scheper et al 1991 [↗](#); Scheper et al 1994 [↗](#)) and inactivation of eIF2 compromises EMCV IRES mediated translation (Welnowska et al 2011 [↗](#); Kwon et al 2016 [↗](#)), indicating EMCV IRES's dependence on canonical ternary complex. Here, we observe a direct interaction of EMCV IRES with ternary complex via tRNA_i, a feature not observed in previously determined HCV and CrPV IRESs-bound ribosomal structures.

As found in canonical 48S PICs (Hashem et al 2013 [↗](#); Brito Querido et al 2024 [↗](#); Petrychenko et al 2024 [↗](#)), eIF2 α -domain 1 is in close contact with ribosomal protein-uS7, and domain 2 with tRNA_i elbow and domain 3 with eIF2 γ in EMCV IRES-48S PIC. The position of eIF2 γ and eIF2 α -domain3 (D3) is distinct from mammalian 48S-closed PICs (PDB Id-6YAN and 7QP7) as we could observe a shift by $\sim 10\text{\AA}$ towards the head of 40S on superimposing the 18S rRNA from EMCV IRES 48S PIC and mammalian late-stage 48S PIC (Fig. 5B [↗](#)). The ternary complex is flexible, and it moves away from ribosomal head towards the body on recognition of authentic start codon as studied in yeast (Villamayor-Belinchón et al 2024 [↗](#)) and human (Petrychenko et al 2024 [↗](#)). This opposite directional shift of the ternary complex in the EMCV IRES-48S PIC is evident in tRNA_i acceptor stem as well (Fig. 5C [↗](#)). This shift could be due to the rigid stem B3 (consisting G-C base pairs), connecting 40S head to GNRA loop, which interacts with the tRNA_i at its elbow and acceptor arm, and the association of eIF2 α -D3 and eIF2 γ with the acceptor arm of tRNA_i orchestrated with the conformational change.

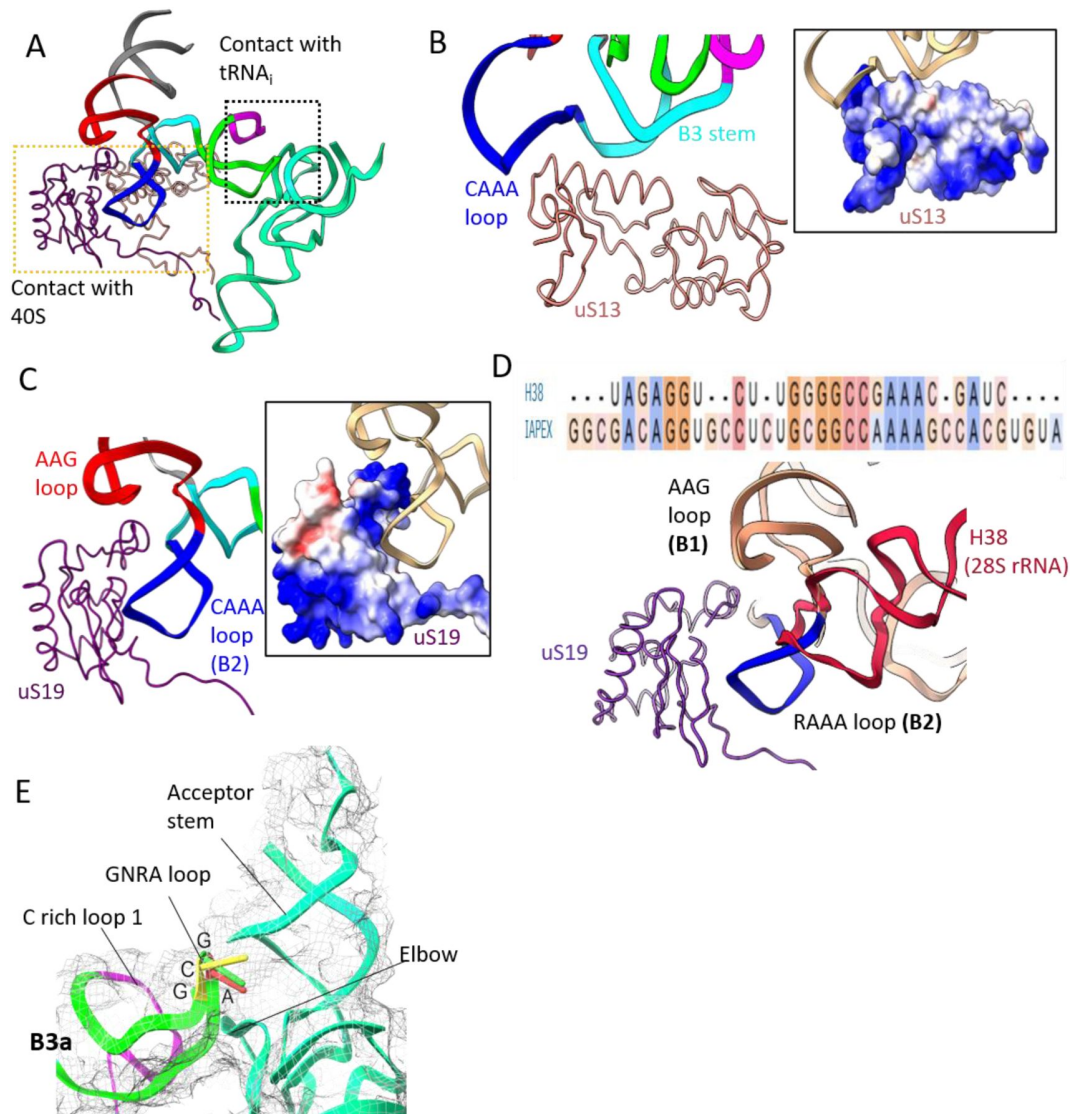


Figure 4

(A) Model showing connections of I domain apex with uS13, uS19, and tRNA_i. **(B)** uS13 interacts with B3 branch or sub-domain of IRES via its alpha helix (100-117 residues). **(C)** Multiple point of contacts between uS19 and I domain motifs-RAAA and AAG. The electrostatic potential map of uS19 suggests that EMCV IRES interacts via ionic interactions with its phosphate backbone. **(D)** Sequence alignment of h38 with I domain apex of EMCV IRES showing sequence identity. Overlapping of uS19 from 80S (PDB Id-4UG0) and EMCV IRES-48S PIC show that the interaction of uS19 with EMCV IRES is similar to interaction with that of h38. **(E)** The fit of GNRA or GCGA stem and its contact with tRNA_i at the elbow region and acceptor stem.

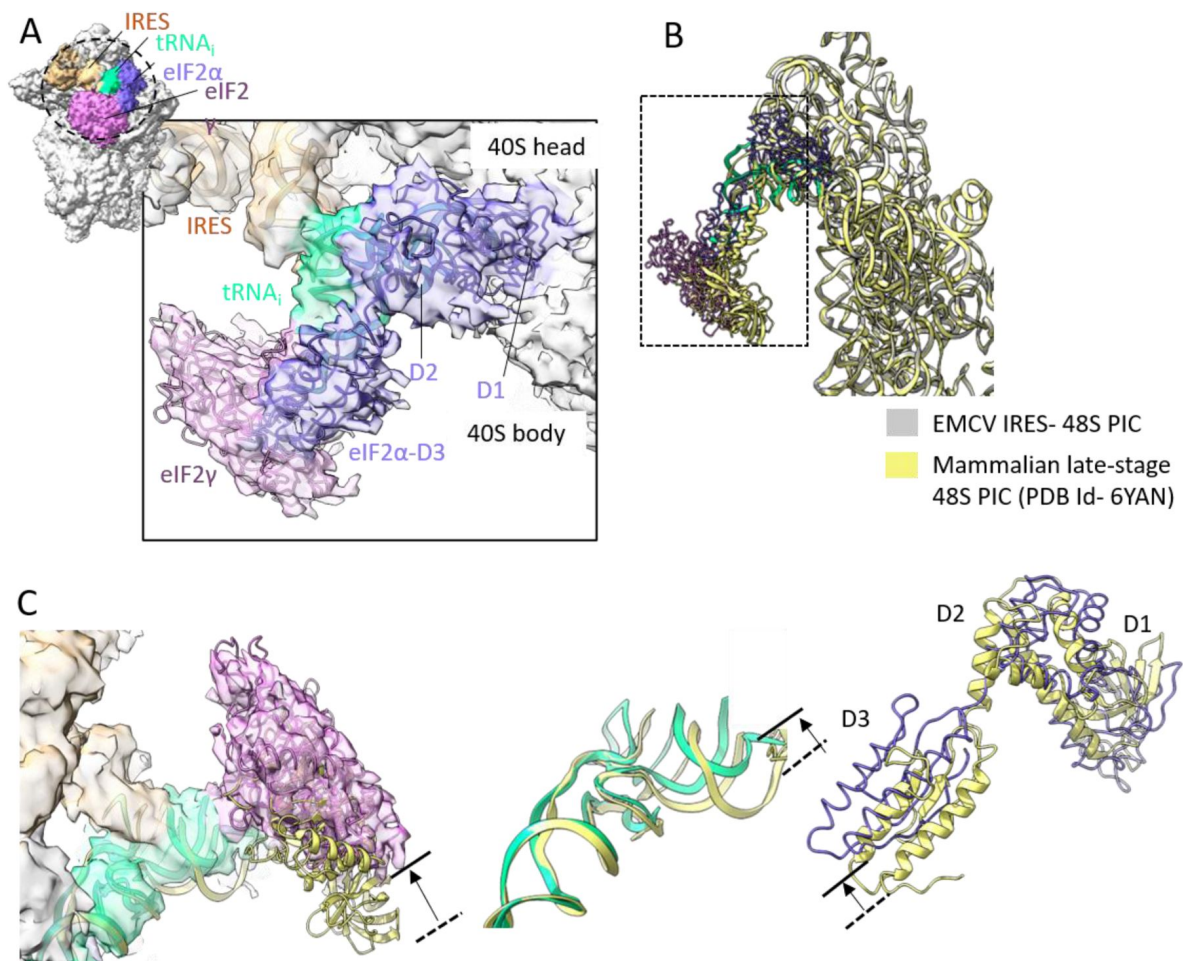


Figure 5

(A) Inter-subunit view of EMCV IRES 48S PIC showing position for ternary complex on map. Fitting of eIF2α and eIF2γ in its corresponding density in Map B1. **(B)** Overlapping EMCV IRES 48S PIC on mammalian late-stage 48S PIC (PDB Id-6YAN), indicates a shift in position of eIF2γ towards 40S head. **(C)** Zoomed view showing position of eIF2γ, tRNA_i, eIF2α from EMCV IRES 48S PIC with respect to position on mammalian late-stage 48S PIC (PDB Id-6YAN). The black arrow indicates the shift in position.

Discussion

The mechanism of IRES recruitment on 40S ribosome varies considerably among different types of IRESs. CrPV intergenic IRES binds to 40S ribosome by mimicking tRNA-mRNA interaction with the help of pseudoknot 1 (Petrov et al 2016 [\[16\]](#); Fernandez et al 2014 [\[17\]](#); Murray et al 2016 [\[18\]](#); Acosta-Reyes et al 2019 [\[19\]](#)), whereas HCV IRESs associates with the solvent side of 40S body by replacing eIF3 with its domain 3 (Spahn et al. 2004 [\[20\]](#); Hashem et al 2013 [\[21\]](#); Niepmann and Gerresheim 2020 [\[22\]](#); Brown et al 2022 [\[23\]](#)). In this study, we capture EMCV IRES in 48S PIC context, interacting with 40S ribosomal head and tRNA_i. Model building suggests that I domain of EMCV IRES interacts with 40S ribosome head and tRNA_i elbow stem (**Fig. 6A** [\[24\]](#)). The similarity in the sequence of h38 of 28S rRNA with the I domain and its ability to interact with uS19-N terminal via RAAA motif (Khatter et al 2015 [\[25\]](#)) suggests a mimicry mechanism adopted by EMCV IRES for its recruitment to the ribosome. Moreover, the conservation of I domain apex sequence and motifs (RAAA, AAG loop, C-rich and GNRA) across Cardioviruses-Theiler's murine Encephalitis virus (TMEV), Vilyuisk human encephalomyelitis virus (VHEV), Theiler's-like rat virus (TRV) and Saffold viruses 1 and 2 (SAFV-1 and SAFV-2) and Aphoviruses (AAG loop is replaced by ACG loop)-Foot and mouth disease virus (Hellen and Wimmer 1995 [\[26\]](#); Liang et al 2008 [\[27\]](#)) suggests that these IRESs might adopt similar strategies for 48S formation (**Fig. 6B-C** [\[28\]](#)). EMCV IRES can directly interact with 40S unlike type 1 IRESs. The I domain of EMCV IRES is similar to domain IV of polioviral IRES (or other type 1 IRESs) in terms of length, secondary structure and conserved motifs (GNRA, C-rich) positioning (**Fig. 6C** [\[29\]](#)), therefore, anticipating a similar interaction with tRNA_i, highlighting a sequestering tendency by competing with cellular mRNAs. However, the lack of RAAA motif in type 1 provides a plausible reason for no direct interaction between type 1 IRES and 40S (Lozano and Martinez-Salas 2015 [\[30\]](#)).

The EMCV IRES does not require scanning, and the start codon (A-834) is directly placed in the P-site, which would eventually place the L domain at the mRNA exit site, preceded by J-K domain that interacts with eIF4G-eIF4A (**Fig. 6A** [\[31\]](#)). Earlier, biochemical studies suggested eIF4G to be positioned close to ES6 of 18S rRNA in EMCV IRES-bound 48S PIC (Yu et al 2011 [\[32\]](#)). The human 48S PIC with a 5' capped mRNA showed a similar location for eIF4G, that is at the mRNA exit site contacting eIF3 (Brito Querido et al 2020 [\[33\]](#); Brito Querido et al 2024 [\[34\]](#)). Locating eIF4F have been challenging due to inherent flexibility associated with the eIF4F complex on mRNA and requires association with eIF3 in canonical 48S context (Brito Querido et al 2020 [\[33\]](#); Brito Querido et al 2024 [\[34\]](#)). However, eIF3-eIF4G interaction is dispensable for EMCV IRES-48S PIC formation, so we do not rule out the possibility that EMCV IRES may dislodge eIF3 from its position on the solvent surface as observed in case of HCV IRES (Hashem et al 2013 [\[21\]](#)). Although, PTB1 serves as an essential ITAF for 48S PIC formation on EMCV IRES, the obtained map shows no distinct density to PTB1. PTB1 binds to H domain and I domain base, and K domain loop (Kafasla et al 2009 [\[35\]](#)) and the flexibility associated with these domains might have hindered capturing of PTB1 in the reported 48S complex (**Fig. 6A** [\[36\]](#)). However, the unassigned extra density at the mRNA entry site could be contributed by PTB1-RRM interacting with 18S rRNA as discussed previously.

The structural studies on type 2 IRESs have been limited be due to their flexible nature. In the cryo-EM map and structural analysis presented here, we could capture a part of EMCV IRES in 48S context (**Fig. 6A** [\[37\]](#)) and fetch a significant understanding of ribosomal recruitment by EMCV IRES and 48S PIC formation. The entrapment of additional factors and a higher resolution would significantly provide more insights into EMCV IRES 48S PIC. The conservation of secondary structures and motifs such as GNRA within the picornaviruses (type 1, 2, and Aichiviral IRESs-Abdullah et al 2023 [\[38\]](#)) provokes common strategies of interaction in the context of 48S PICs in the Picornaviridae family.

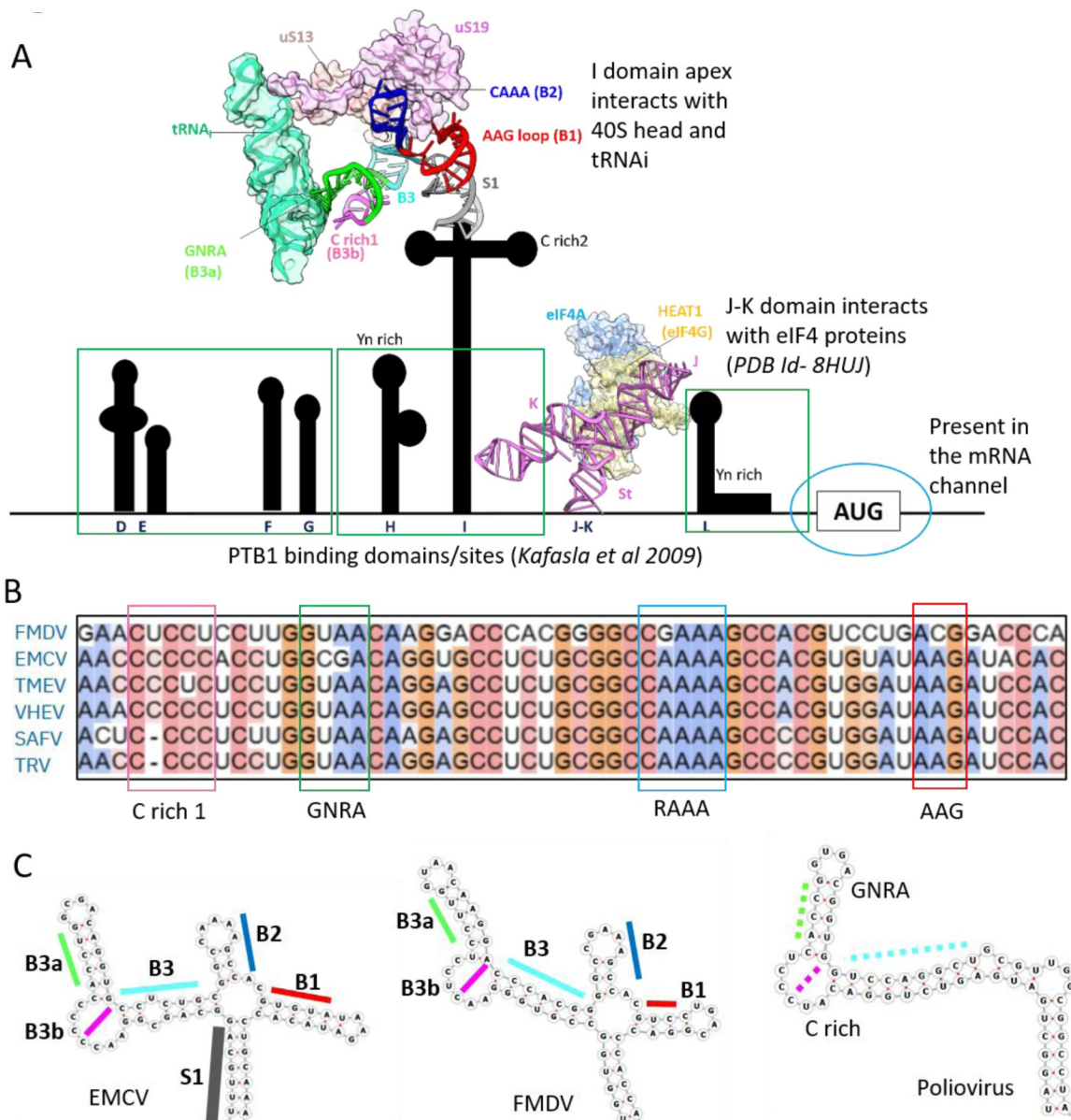


Figure 6

(A) EMCV IRES secondary structure depicting the role or binding partners for each domain in context to translation initiation. The known tertiary structure of EMCV IRES and its binding partners are depicted. The domains responsible for binding PTB1 are boxed. (B) Conservation of I domain apex sequence and secondary structure across type 2 IRES family. (C) Comparison of secondary structure of EMCV and FMDV IRES to that of Poliovirus IRES (type 1).

Materials and methods

1. Plasmid constructs and Molecular Cloning

EMCV IRES 905 was obtained from EMCV-L plasmid-EMCV IRES (nt 280–905) into pCR2.1 and inserted in pcDNA3.1 using BamH1 and Xba1 restriction sites. Polypyrimidine tract binding protein 1 was cloned from HEK293 cDNA and 3C protease site (LEVLFQGP) was inserted at the N-terminal and inserted in pET28a in between BamH1 and HindIII restriction sites, retaining the N-terminal 6X Histidine tag. Sequence of oligos or primers used is listed in Sup. [Table 1](#).

2. Protein overexpression and purification

PTB1 (Histidine tag-3Cprotease site-PTB1 gene) was overexpressed using 0.5mM IPTG in *E. coli* BL21 cells at 30°C, 120 rpm for 4 hours in Luria broth. 2l harvested culture was lysed using Buffer N1 (20 mM HEPES pH-7.4, 300 mM KCl, 2mM MgCl₂, 10 % Glycerol, 10mM Imidazole, 5mM β-Mercaptoethanol, 0.05% Triton-X-100, 2mM PMSF) and sonicated at 18% amplitude, 10s ON and 20s OFF pulses, 30 cycles and centrifuges at 20000 rpm for 20 minutes and the supernatant was loaded on Ni-NTA column and eluted using a gradient of N1 to N2 buffer (20 mM HEPES pH-7.4, 300 mM KCl, 2mM MgCl₂, 10 % Glycerol, 500mM Imidazole, 5mM β-Mercaptoethanol). PTB1 was eluted at 250 mM Imidazole concentration. The eluant fractions (diluted to 100 mM KCl) were further loaded on Heparin column and eluted by applying a gradient of 100 mM to 1000 mM KCl. The protein was eluted at 250 mM KCl concentration. The eluant fractions were stored at -80°C after Size Exclusion Chromatography in Buffer S (20 mM HEPES pH-7.4, 200 mM KCl, 2mM MgCl₂, 5 % Glycerol, 1mM DTT) at 2.5 mg/ml concentration.

3C protease was overexpressed using 0.5mM IPTG in *E. coli* BL21 cells at 30°C, 120 rpm in Luria broth. 2l harvested culture was lysed using Buffer N11 (20 mM HEPES pH-7.4, 200 mM KCl, 2mM MgCl₂, 10 % Glycerol, 10mM Imidazole, 5mM β-Mercaptoethanol, 0.05% Triton-X-100, 2mM PMSF) and sonicated at 18% amplitude, 10s ON and 20s OFF pulses, 30 cycles and centrifuges at 20000 rpm for 20 minutes and the supernatant was loaded on Ni-NTA column and eluted using a gradient of N11 to N22 buffer (20 mM HEPES pH-7.4, 200 mM KCl, 2mM MgCl₂, 10 % Glycerol, 500mM Imidazole, 5mM β-Mercaptoethanol). 3C protease was eluted at 250 mM Imidazole concentration. The eluant fractions were subjected to Size Exclusion Chromatography in Buffer S (20 mM HEPES pH-7.4, 100 mM KOAc, 2mM MgCl₂, 10 % Glycerol, 1mM DTT) and peak fractions were concentrated using 3kDa cutoff centricon and stored at 2.5 mg/ml concentration.

3. In vitro transcription of EMCV IRES

EMCV IRES 905-pcDNA3.1 was linearized using Xba1 restriction enzyme and transcribed using Promega-RiboMAX™ Large Scale RNA Production Systems as per manufacturer's protocol. 1 µg of linearized plasmid yielded 87 µg of RNA after DNase treatment and RNA cleanup (RNA clean up kit-NEB).

4. Assembly of EMCV IRES 48S PIC using Talon Affinity Chromatography

12 µg of IRES was heated at 95°C to dissolve any secondary structures acquired and refolded using Buffer R (20 mM HEPES pH-7.4, 150 mM KOAc, 2mM MgCl₂, 2mM β-Mercaptoethanol, 0.5mM Spermidine) at 37°C for 5 mins and PTB1 (1:2 = IRES: PTB1) is added and further incubated at 30°C for 5 mins. Simultaneously, Rabbit Reticulocyte Lysate (RRL) (Promega) was incubated at 30°C with ATP, Amino acid mix minus Leucine, Murine RNase Inhibitor for 5 minutes and then mixed with IRES-PTB1 vial with instant addition of 6mM GMP-PnP per reaction (100 µl) and incubated at 30°C for 8 minutes, followed by ice incubation. The reaction was loaded on Talon beads equilibrated

with Buffer A (20 mM HEPES pH-7.4, 150 mM KOAc, 2mM MgCl₂, 4 % Glycerol, 5mM Imidazole, 2mM β-Mercaptoethanol). After recommended passes, reloading and incubation, the flow through was collected and the beads were washed with Buffer A until A260 attains a baseline value (~0), following which 3C protease in Buffer A was added to the beads and left for overnight incubation. Fractions were eluted using Buffer A (400 μl each in 10 vials) and subjected to analysis such as A260 measurements, and agarose gel electrophoresis. Samples having the RNA bands were pelleted using an 1ml Sucrose cushion (20 mM HEPES pH-7.4, 150 mM KOAc, 2mM MgCl₂, 30% Sucrose, 1mM DTT) in SW60 tubes and centrifuged at 50,000 rpm for 10 hours at 4°C. The pellet was resuspended in 20μl buffer R (+0.2 mM Spermidine) and used for Cryo-EM grid preparation (No crosslinker was used to avoid artifacts).

5. Negative stain analysis and Mass Spectrometry

Final sample was diluted 10 times using Buffer R and applied on 400-mesh Cu TEM grids which were freshly glow-discharged (negative polarity) for 30 s in GloQube glow-discharge system, stained using 1% Uranyl acetate solution and analysed using microscope at 57000 X magnification. Furthermore, the sample was digested by trypsin and subjected to NanoOrbitrap analysis for identification of proteins in the complex.

6. Cryo-EM sample preparation

3 μl of resuspended pellet (2.96 A260) was applied on glow discharged Quantifoil R 1.2/1.3 300 mesh 2nm Carbon Coated Grid and Blotted using 8s and 8.5s Blot time, zero Blot force at 16°C and 100% Humidity and plunged into liquid ethane. Cryo-EM data were collected on Talos Arctica transmission electron microscope equipped with a FEG at 200 kV (Thermo Fisher Scientific). All data were collected using a Gatan K2 Summit Direct Detector at a nominal magnification of 36,000 X and a pixel size of 1.17 Å with a total electron dose of 55 e-/Å² fractionated over 20 frame movies with a dose rate of ~2.5 e-/Å²/frame.

7. Data Processing

Micrographs were collected and processed using CryoSPARC v3.3 (Punjani et al 2017 [\[1\]](#)). The micrographs were patch motion corrected and CTF was estimated. Using a blob picker with an average diameter of 300 Å particles were picked and extracted. The extracted particles were subjected to multiple rounds of 2D classification. Final 2D classes showing promising ribosome 2D features were selected for *ab initio* reconstruction. The junk was discarded, and the 237054 good particles were classified into 2 classes using a mask around tRNA_i. We obtained 2 classes-Empty 40S (1.25L particles-Class 1 or Map A), and 40S with tRNA bound (1.11L particles-Class 2). Class 2 was further classified into 2 classes using a 3D mask around the IRES density and subjected to non-uniform refinement with global CTF refinement to yield Map B, having tRNA_i and IRES density (55k particles). This was subjected to Non-uniform Refinement (Punjani et al 2020 [\[2\]](#)) with global CTF refinement. Map B class was further classified using a mask around eIF2α and eIF2γ to yield 2 classes-40S-tRNA_i-IRES-eIF2 (28k particles-Map B1) and 40S-tRNA_i-IRES (26k particles). The obtained maps were then subjected to model building and refinement. The local resolution for the obtained maps were estimated using Phenix (Liebschner et al 2019 [\[3\]](#)).

8. Map analysis and Model building

To the obtained maps, 40S ribosome (PDB Id-6YAN) was fitted using UCSF Chimera. The 40S head (Head proteins + 18S rRNA (1197-1688) + eS17) and 40S body (Body proteins + 18S rRNA (1-1196; 1689-1870)) was fitted to the obtained maps separately and subjected to Rigid Body Fit and Real Space Refinement using PHENIX (Liebschner et al 2019 [\[3\]](#)). The models were merged using Coot (Emsley et al 2010 [\[4\]](#)) and subjected to Real Space Refinement. tRNA_i and mRNA models were taken from PDB Id-8OZO and fitted to the maps-B and B1 in Chimera. eIF2α and eIF2γ from PDB Id-8OZO were rigid-body fitted to Map B1 and then mutated as per Rabbit eIF2 protein sequence (NCBI Reference Sequence-XP_002719561.1; XP_051683593.1) and subjected to a final Real Space

Refinement. The IRES I domain apex model was predicted from AlphaFold3 (Abramson et al 2024 [↗](#)) and the helical sub-domains were dismantled and fitted according to best fit using Chimera, followed by chain joining in Coot with manual Real Space Refinement (Afonine et al 2018 [↗](#)). The geometry was corrected using Geometry minimization tool in Phenix with rounds of Real Space Refinement. H domain model was predicted using AlphaFold3. The final model yielded 40S-EMCV IRES-tRNA_i (Map B) and 40S-EMCV IRES-tRNA_i-eIF2αY (Map B1). All the figures were made using ChimeraX (Pettersen et al 2021 [↗](#)). The models were real space refined using Phenix and the Fourier Shell Correlation for 'Map to Model' for each was determined at 0.5 FSC (Sup. Fig. 2A-C; **Table 1** [↗](#)).

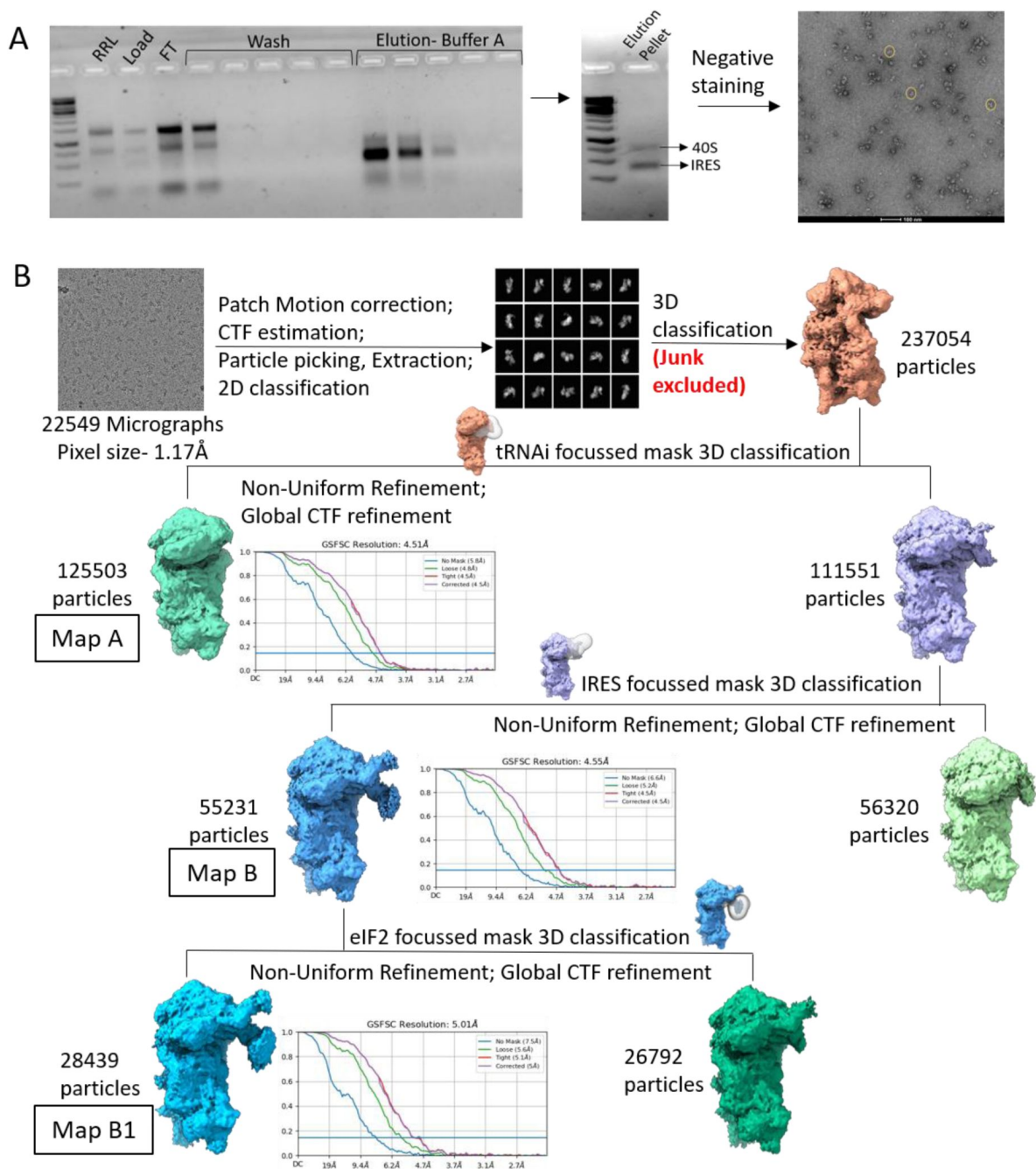
9. Secondary Structure Determination and Multiple sequence alignments

The secondary structures for I domain apex was obtained from RNAfold (Gruber et al 2008 [↗](#)). Multiple sequence alignments were performed using Clustal Omega (Sievers et al 2021 [↗](#)). Sequence accession number for various sequence used-EMCV (NC_001479.1), FMDV (NC_039210.1), TMEV (DQ401688.1), TRV (AB090161.1), VHEV (M80888.1), SAFV (FM207487.1), Poliovirus (NC_002058.3).

Data availability

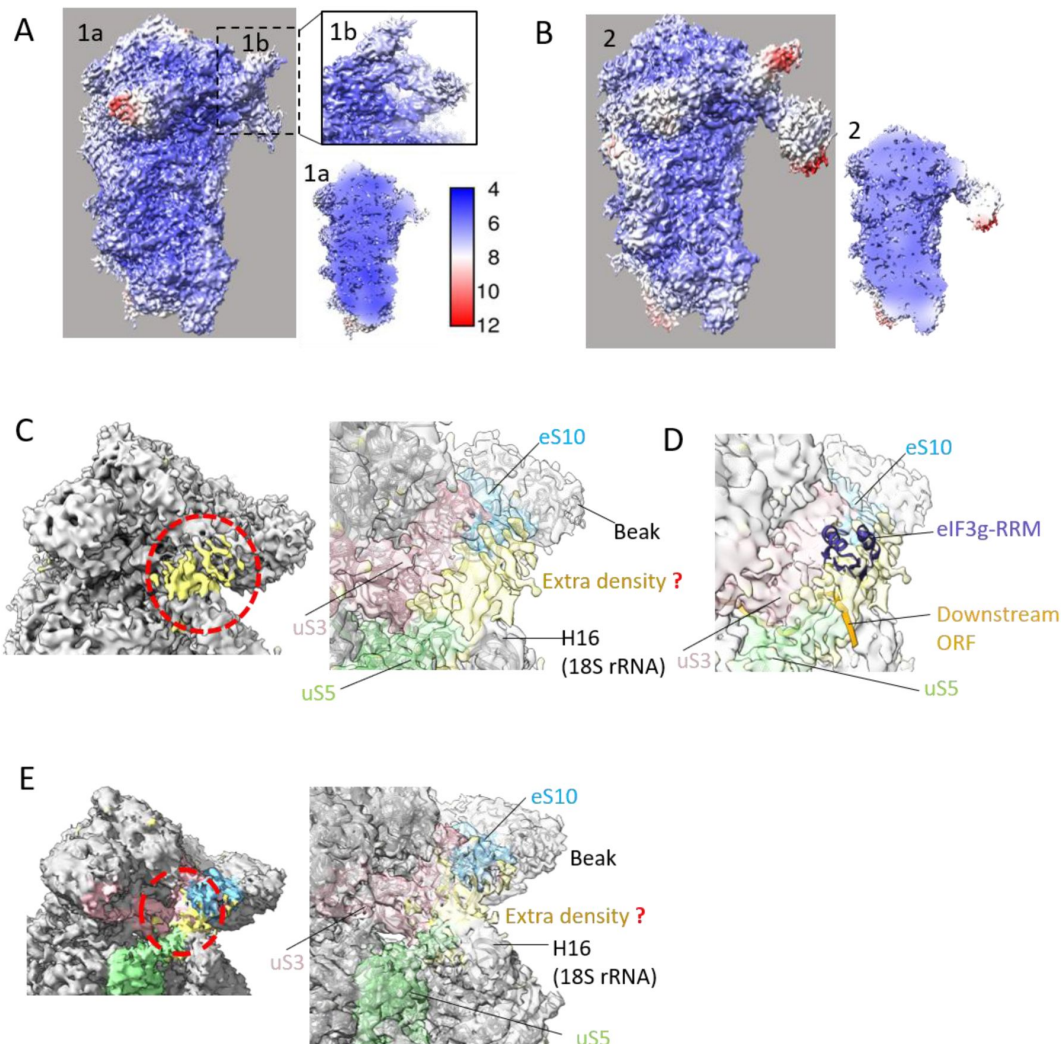
Maps and atomic coordinates of the 40S ribosome (Map A), 40S ribosome-EMCV IRES-tRNA_i (Map B) and 40S ribosome-EMCV IRES-ternary complex (Map B1) have been deposited in the EMDB database and in the PDB database.

Supplementary figures and table



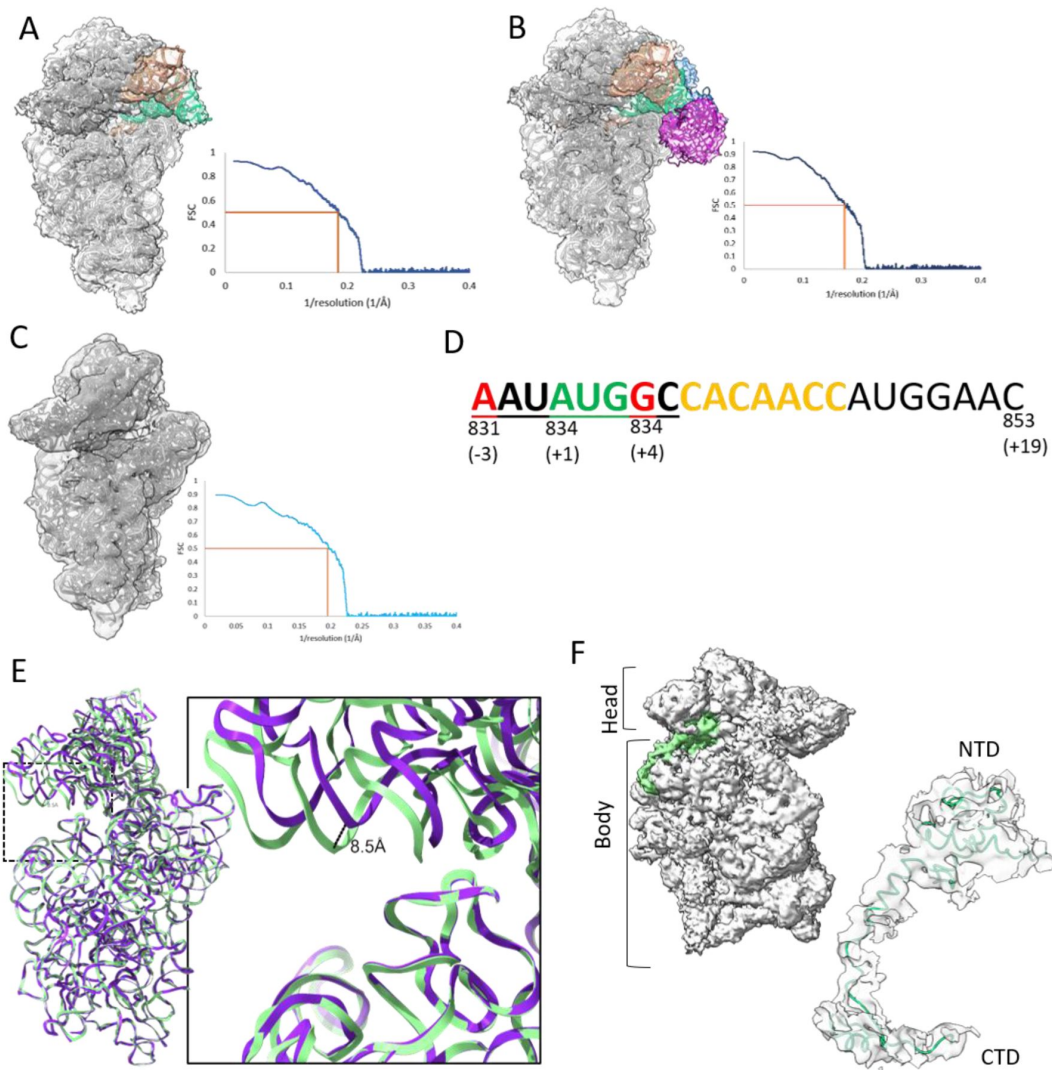
Supplementary Figure 1.1

(A) Isolation of EMCV IRES-48S PIC from Talon Affinity Chromatography-Talon Affinity chromatography profile and analysis of Elution fractions-pelleted by using 30% cushion on 1.5% Agarose gel, followed by Uranyl acetate staining of elution pellet and observation under transmission electron microscope at 57k X. **(B)** Processing of Cryo-EM data and features of obtained maps using CryoSPARCv4.3.0 and GSFSC resolution estimates for different maps obtained.



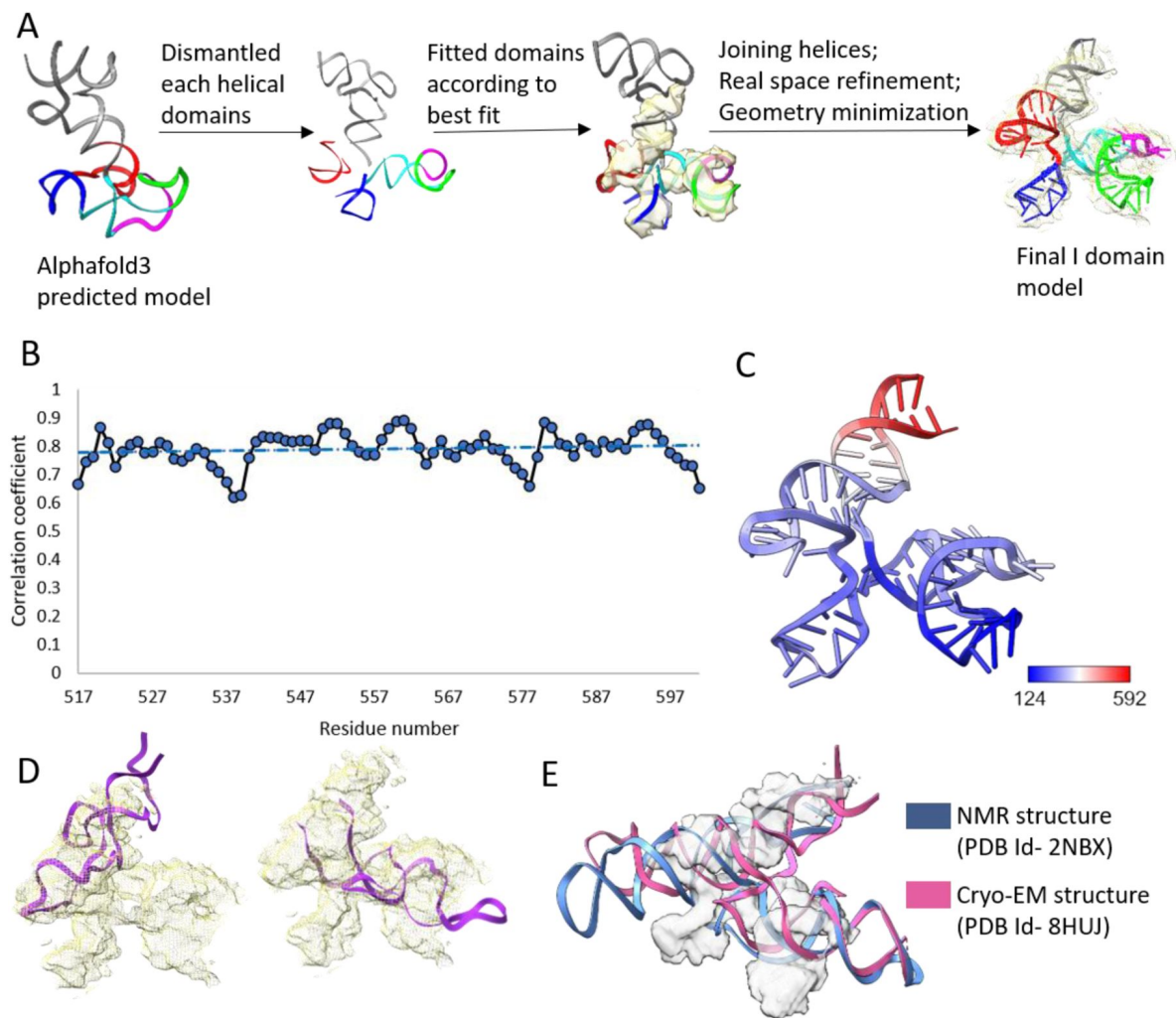
Supplementary Figure 1.2

(A) Local resolution estimation of Map B. (right) Surface view; (right-top) zoomed view on IRES density; (left-bottom) cross-section of Map B to show resolution at the core of the ribosome. **(B)** Local resolution estimation of Map B1. (right) Surface view; (left-bottom) cross-section of Map B1 to show resolution at the core of ribosome. **(C)** Extra density at the mRNA entry site of Map B/B1. Ribosomal proteins and RNA in contact with the unassigned extra density. **(D)** Superimposition of EMCV IRES-48S PIC on human 48S PIC (PDB Id-8OZ0) shows the extra density coincides with the location of eIF3g-RRM and downstream mRNA. **(E)** Extra density at the mRNA entry site of Map A. Ribosomal proteins and RNA in contact with the unassigned extra density.



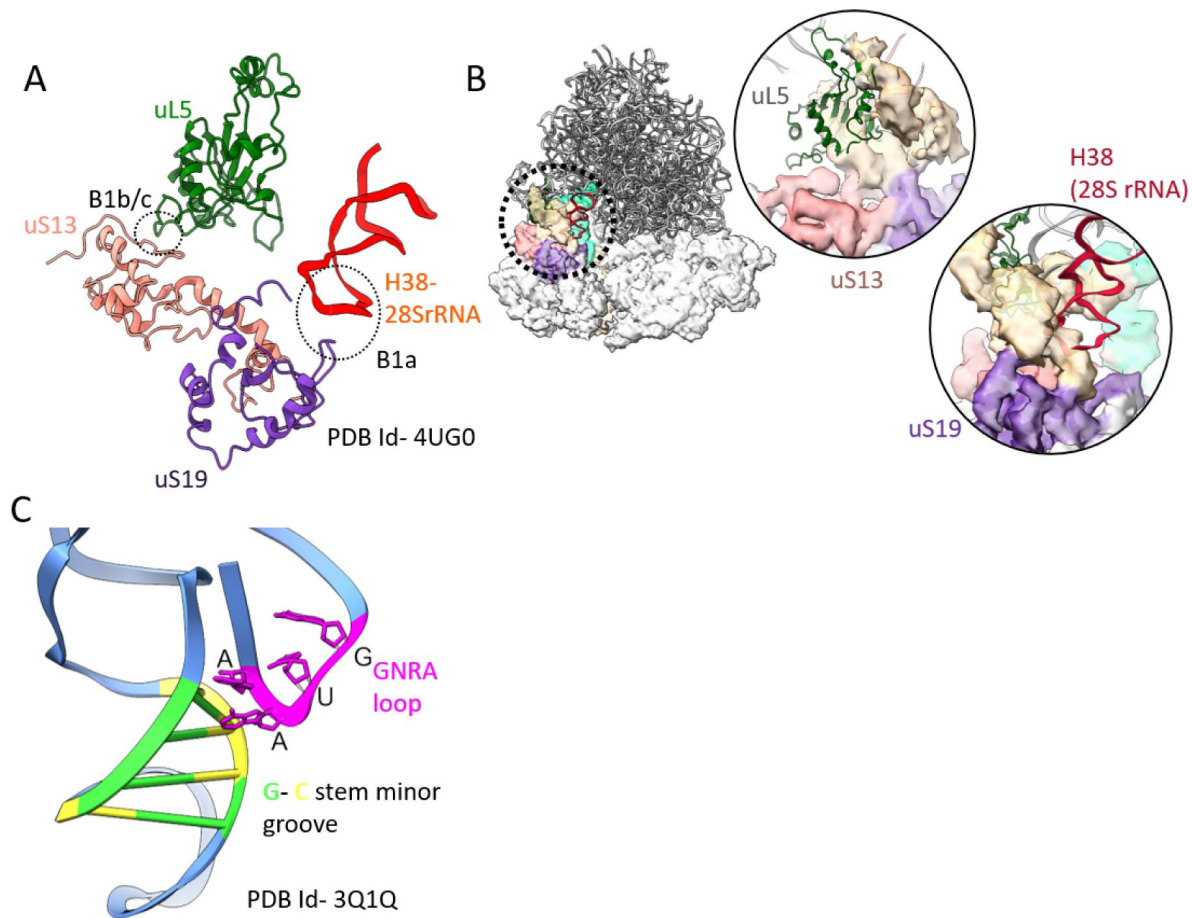
Supplementary Figure 2

(A) From Map B to model (EMCV IRES-48S PIC). Obtained Model showing 40S ribosome, RNA in channel, initiator tRNA, IRES domain. (right) Fourier Shell Correlation (FSC) of Map to Model fit. **(B)** From Map B1 to model (EMCV IRES-48S PIC). Obtained Model showing 40S ribosome, RNA in channel, initiator tRNA, IRES domain, eIF2α. (right) FSC of Map to Model fit. **(C)** From Map A to model by fitting of 40S ribosome without any factors. (right) FSC of Map to Model fit. **(D)** The sequence of EMCV IRES mRNA in the channel is provided. Residues in orange colour has no density in the obtained map B. **(E)** Comparing the state of 18S rRNA in Map A and Map B. **(F)** eS17-NTD position in 40S ribosome. (right) Fitting of eS17 in EMCV IRES-48S PIC map.



Supplementary Figure 3

(A) Process of fitting I domain apex from AlphaFold3 generated model. **(B)** Correlation coefficient per residue in I domain apex for fit in the density map. **(C)** B-factor for each residue in the final model. The range is depicted in the colour key where blue is low, and red is high. **(D)** Possibility of fitting H domain in the IRES density. H domain model was generated by AlphaFold3 (ptm= 0.31). **(E)** Possibility of fitting J-K domain (PDB-8HUJ) in density. The J-K domain is way too long to fit in map and does not account for certain region. The extending loops would clash with 40S ribosome and tRNA_i.



Supplementary Figure 4

(A) Interaction of uS19 and uS13 in 40S subunit with 60S subunit-Helix 38 and uL5 to form inter-subunit bridges-B1a and B1b/c, respectively in 80S (PDB Id-4UG0). **(B)** Estimation of clash with IRES domain-I during 60S joining to form 80S-elongation competent complex (PDB Id-4UG0). uL5 and helix 38 of 28S rRNA would clash with the IRES I domain. (left) Superimposition of 80S ribosome on EMCV IRES-48S PIC. **(C)** Reported GNRA (GUGA) interaction with the minor groove formed by C-G stem in bacterial RNaseP (PDB Id-3Q1Q).

Oligo name	Sequence
PTB1_Fwd	CGGGATCCATGGACGGCATTGTCCCAGA
PTB1_Rev	CCCAAGCTTCTAGATGGTGGACTTGGAG
PTB1_3C_Fwd	CGGGATCCCTGGAGGTGCTCTTCCAGGGCCCTGGCGGCTCCATGGACGGCATTGTCCC AG
PTB1_Stop_Rev	CCCAAGCTTCTAGATGGTGGACTTGGAGAAGG
EMCV_IRES_905 -Fwd	CGGGATCCCCCCTAACGTTACTGGCC
EMCV_IRES_905 -Rev	GCTCTAGAGCAGAGCATTTTGGGCATTCTCAAAAG

Supplementary Table 1

List of Oligos or primers used for molecular cloning

Acknowledgements

EMCV IRES (nt 280–905) into pCR2.1 was a kind gift from Dr. Bruno Sargueil, CNRS UMR8015, Université Paris Descartes, France and 3C protease-pET24 was a kind gift from Prof. Raghavan Varadarajan, Indian Institute of Science (IISc), India. We thank various central facilities, namely Cryo-EM, Mass Spectrometry and Computational cluster at the Division of Biological Sciences, IISc for support. DD acknowledges the DBT-JRF Programme for fellowship. The authors acknowledge the DST-FIST support to the department. This work was supported by the Intermediate Fellowship from DBT-Wellcome Trust India Alliance to TH (IA/I/17/2/503313).

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Editors

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Reviewer #1 (Public review):

Summary:

The authors have studied how a virus (EMCV) uses its RNA (Type 2 IRES) to hijack the host's protein-making machinery. They use cryo-EM to extract structural information about the recruitment of viral Type 2 IRES to ribosomal pre-IC. The authors propose a novel interaction mechanism in which the EMCV Type 2 IRES mimics 28S rRNA and interacts with ribosomal proteins and initiator tRNA (tRNAi).

Strengths:

- (1) Getting structural insights about the Type 2 IRES-based initiation is novel.
- (2) The study allows a good comparison of other IRES-based initiation systems.
- (3) The manuscript is well-written and clearly explains the background, methods, and results.

Weaknesses:

- (1) The main weakness of the work is the low resolution of the structure. This limits the possibility of data interpretation at the molecular level.

However, despite the moderate resolution of the cryo-EM reconstructions, the model fits well into the density. The analysis of the EMCV IRES-48S PIC structure is thorough and includes meaningful comparisons to previously published structures (e.g., PDB IDs - 7QP6 and 7QP7). These comparisons showed that Map B1 represents a closed conformation, in contrast to Map A in the open state (Figure 2). Additionally, the proposed 28S rRNA mimicry strategy supported by structural superposition with the 80S ribosome and sequence similarity between the I domain of the IRES and the h38 region of 28S rRNA (Fig. 4) is well-justified.

- (2) The lack of experimental validation of the functional importance of regions like the GNRA and RAAA loops is another limitation of this study.

- (3) Minor modifications related to data processing and biochemical studies will further validate and strengthen the findings.

a) In the cryo-EM data section, the authors should include an image showing rejected particles during 2D classification. This would help readers understand why, despite having over 22k micrographs with sufficient particle distribution and good contrast, only a smaller number of particles were used in the final reconstruction. Additionally, employing map-sharpening tools such as Ewald sphere correction, Bayesian polishing, or reference-based motion correction might further improve the quality of the maps. Targeting high-resolution structures would be particularly informative.

b) The strategic modelling of different IRES domains into the density, particularly the domain into the region above the 40S head, is appreciable. However, providing the full RNA tertiary structure (RNAfold) of the EMCV IRES (nucleotides 280-905) would better explain the logic behind the model building and its molecular interpretation.

c) Although the authors compare their findings with other types of IRESs (Types 1, 3, and 4), there is no experimental validation of the functional importance of regions like the GNRA and RAAA loops. Including luciferase-based assays or mutational studies of these regions for validation of structural interpretations is strongly recommended.

<https://doi.org/10.7554/eLife.107788.1.sa3>

Reviewer #2 (Public review):

Summary:

The field of protein translation has long sought the structure of a Type 2 Internal Ribosome Entry Site (IRES). In this work, Das and Hussain pair cryo-EM with algorithmic RNA structure prediction to present a structure of the Type 2 IRES found in Encephalomyocarditis virus (EMCV). Using medium to low resolution cryo-EM maps, they resolve the overall shape of a critical domain of this Type 2 IRES. They use algorithmic RNA prediction to model this domain onto their maps and attempt to explain previous results using this model.

Strengths:

- (1) This study reveals a previously unknown/unseen binding modality used by IRESes: a direct interaction of the IRES with the initiator tRNA.
- (2) Use of an IRES-associated factor to assemble and pull down an IRES bound to the small subunit of the ribosome from cellular extracts is innovative.
- (3) Algorithmic modeling of RNA structure to complement medium to low resolution cryo-EM maps, as employed here, can be implemented for other RNA structures.

Weaknesses:

- (1) Maps at the resolution presented prevent unambiguous modelling of the EMCV-IRES. This, combined with the lack of any biochemical data, calls into question any inferences made at the level of individual nucleotides, such as the GNRA loop and CAAA loop (Figure 4).
- (2) The EMCV IRES contains an upstream AUG at position 826, where the PIC can assemble (Pestova et al 1996; PMID 8943341). It is unclear if this start codon was mutated in this study. If it were not mutated, placement of AUG-834 over AUG-826 in the P-site is unexplained.
- (3) The claims the authors make about (i) the general overall shape and binding site of the IRES, (ii) its gross interaction with the two ribosomal proteins, (iii) the P-in state of the 48S, (iv) the rearrangement of the ternary complex are all warranted. Their claims about individual nucleotides or smaller stretches of the IRES-without any supporting biochemical data-is not warranted by the data.

<https://doi.org/10.7554/eLife.107788.1.sa2>

Reviewer #3 (Public review):

Summary:

Type II IRES, such as those from encephalomyocarditis virus (EMCV) and foot-and-mouth disease virus (FMDV), mediate cap-independent translation initiation by using the full complement of eukaryotic initiation factors (eIFs), except the cap-binding protein eIF4E. The molecular details of how IRES type II interacts with the ribosome and initiation factors to promote recruitment have remained unclear. Das and Hussain used cryo-electron microscopy to determine the structure of a translation initiation complex assembled on the EMCV IRES. The structure reveals a direct interaction between the IRES and the 40S ribosomal subunit, offering mechanistic insight into how type II IRES elements recruit the ribosome.

Strengths:

The structure reveals a direct interaction between the IRES and the 40S ribosomal subunit, offering mechanistic insight into how type II IRES elements recruit the ribosome.

Weaknesses:

While this reviewer acknowledges the technical challenges inherent in determining the structure of such a highly flexible complex, the overall resolution remains insufficient to fully support the authors' conclusions, particularly given that cryo-EM is the sole experimental approach presented in the manuscript.

The study is biologically significant; however, the authors should improve the resolution or include complementary biochemical validation.

<https://doi.org/10.7554/eLife.107788.1.sa1>

Author response:

Reviewer #1 (Public review):

Summary:

The authors have studied how a virus (EMCV) uses its RNA (Type 2 IRES) to hijack the host's protein-making machinery. They use cryo-EM to extract structural information about the recruitment of viral Type 2 IRES to ribosomal pre-IC. The authors propose a novel interaction mechanism in which the EMCV Type 2 IRES mimics 28S rRNA and interacts with ribosomal proteins and initiator tRNA (tRNAⁱ).

Strengths:

- (1) Getting structural insights about the Type 2 IRES-based initiation is novel.*
- (2) The study allows a good comparison of other IRES-based initiation systems.*
- (3) The manuscript is well-written and clearly explains the background, methods, and results.*

We thank Reviewer 1 for appreciating our efforts and finding structural insights about the type 2 IRES-based initiation presented in this study as novel.

Weaknesses:

- (1) The main weakness of the work is the low resolution of the structure. This limits the possibility of data interpretation at the molecular level.*

However, despite the moderate resolution of the cryo-EM reconstructions, the model fits well into the density. The analysis of the EMCV IRES-48S PIC structure is thorough and includes meaningful comparisons to previously published structures (e.g., PDB IDs - 7QP6 and 7QP7). These comparisons showed that Map B1 represents a closed conformation, in contrast to Map A in the open state (Figure 2). Additionally, the proposed 28S rRNA mimicry strategy supported by structural superposition with the 80S ribosome and sequence similarity between the I domain of the IRES and the h38 region of 28S rRNA (Fig. 4) is well justified.

We agree that the low resolution of the map has compromised the data interpretation at the molecular level, and we thank the reviewer for appreciating our findings at this resolution. Due to the compromise in resolution, we have reported findings related to stretches or regions such as loops and stems, rather than individual nucleotides and interactions.

(2) The lack of experimental validation of the functional importance of regions like the GNRA and RAAA loops is another limitation of this study.

We agree with the lack of any additional experiments other than Cryo-EM for probing the importance of regions such as GNRA and RAAA loops in this study. However, we have cited earlier reports that demonstrate the importance of these regions for overall IRES activity. The essentiality of RAAA loop for type 2 IRES was demonstrated in earlier report López de Quinto and Martínez-Salas, 1997 (Cited in manuscript). Further, the conservation of this loop across the type 2 IRES family adds to the importance of this loop (Manuscript Figure 6B). This loop and its flanking G-C stem are similar to h38 of 28S rRNA, and it appears that RAAA loop adopts a mimicry mechanism to interact with the 40S ribosomal protein- uS19, thus highlighting its importance for interaction with 40S. Experiments destabilising the G-C stem also compromise IRES activity, as shown in the case of FMDV IRES (Fernández et al 2011). Previous studies related to the mutation of the GNRA or GCGA loop in EMCV IRES have shown a deficiency in IRES activity (Roberts and Belsham, 1997; Robertson et al 1999), suggesting the importance of these regions in the viral IRES biology, and these reports are cited in the manuscript. Not only EMCV IRES, but mutation in the GUAA (representative of GNRA) loop of FMDV IRES also showed significant reduction in IRES activity (López de Quinto and Martínez-Salas, 1997). In our study, we observe that GCGA loop interacts with tRNA_i in EMCV IRES-48S PIC, thus implicating the importance of this loop. Moreover, incubation of FMDV IRES with 40S ribosomes has shown a decrease in SHAPE reactivity in domain 3 apex (position 170- 200 nucleotides) (Lozano et al 2018), which corresponds to EMCV IRES domain I apex. Further, we will attempt to address the concern of lack of experimental validation of GNRA and RAAA loops by performing biochemical assays.

(3) Minor modifications related to data processing and biochemical studies will further validate and strengthen the findings.

a) In the cryo-EM data section, the authors should include an image showing rejected particles during 2D classification. This would help readers understand why, despite having over 22k micrographs with sufficient particle distribution and good contrast, only a smaller number of particles were used in the final reconstruction. Additionally, employing mapsharpening tools such as Ewald sphere correction, Bayesian polishing, or reference-based motion correction might further improve the quality of the maps. Targeting high-resolution structures would be particularly informative.

We thank the reviewer for the suggestions, and we would employ suggested processes that may help improve the quality of the maps further. We will include image for rejected 2D classes in the revised manuscript. We agree with the Reviewer's query related to the substantial number of micrographs and smaller number of particles for the final

reconstruction. The total number of micrographs is the summation of multiple datasets, prepared and collected at various times. Among these, around 8000 micrographs have extremely poor particle number and distribution. As a result, the number of particles per micrograph is heterogeneous in the compiled dataset. We obtained only 237054 ‘good particles’ after multiple rounds of 2D & 3D classifications, and the final reconstruction has 28439 particles (~12%). This class was obtained after masked classification for IRES and ternary complex density. Hence, only the particles that show the best density for both IRES and ternary complex are used for reconstructing this map. Another set of particles that have only a portion of IRES and tRNA but NO density for eIF2 forms another map (26792 particles, 11.3%). Thus, we obtained a total of 55231 particles (23.3%) with IRES density.

b) The strategic modelling of different IRES domains into the density, particularly the domain into the region above the 40S head, is appreciable. However, providing the full RNA tertiary structure (RNAfold) of the EMCV IRES (nucleotides 280-905) would better explain the logic behind the model building and its molecular interpretation.

We thank the reviewer for appreciating the modelling of the domain I apex in the cryo-EM density. We tried to predict the full tertiary structure of the IRES, however, inclusion of the full-length sequence from 280-905 gave models of extremely low confidence, and few domains do not abide by the secondary structure of EMCV IRES as reported in Duke et al 1992. Hence, we used individual domains of EMCV IRES and predicted the tertiary structure independent of other IRES domains. Furthermore, 3D models of FMDV IRES domains 2, 3, and 4 (corresponding to EMCV IRES domains- H, I, and J-K) were predicted from SHAPE reactivity values and RNAComposer server (Figure 3 in Lozano et al 2018). The predicted architecture of domain 3 apex (FMDV IRES) coincides with our I domain apex model (EMCV IRES).

c) Although the authors compare their findings with other types of IRESs (Types 1, 3, and 4), there is no experimental validation of the functional importance of regions like the GNRA and RAAA loops. Including luciferase-based assays or mutational studies of these regions for validation of structural interpretations is strongly recommended.

We have discussed the possibility of how the other IRESs, such as type 1 and type 5 (Aichi virus), might use similar strategies as EMCV IRES to assemble the 48S PIC, given the similarity in the motif sequence and position across the viral IRESs. Like EMCV IRES, the type 1 IRES (e.g. Poliovirus, Coxsackie virus) also harbours the GNRA loop, preceded by a C-rich loop at its longest domain, known for long-range RNA-RNA interactions. The segment harbouring GNRA loop is highly conserved across the type 1 family of IRESs (Kim et al 2015). The Aichi viral IRES (type 5) harbours a GNRA loop in its longest domain, which is domain J. Deletion of the GNRA loop has compromised the IRES activity; however, substitution mutations in this region either elevated the IRES activity or it remained unaltered (Yu et al 2011). We have hypothesized that these IRESs (type 1 and type 5) might use the GNRA motifs in their longest domain (domain IV in type 1, and domain J in type 5) similar to that of EMCV IRES, where GNRA is present in the longest domain (I) and preceded by a C-rich loop. Thus, GNRA can potentially mediate long-range interactions with tRNA_i as all these IRESs require eIF2-ternary complex for the formation of 48S PIC. Parallely, like EMCV IRES, type 1 and type 5 IRESs also have similar placement of GNRA motif-containing domain before the eIF4G-binding domain (domain J-K in EMCV IRES, domain V in poliovirus, domain K in Aichi virus). Hence, we suggest the possibility of a similar strategy by these IRESs to interact with tRNA_i during the formation of 48S PIC.

Reviewer #2 (Public review):

Summary:

The field of protein translation has long sought the structure of a Type 2 Internal Ribosome Entry Site (IRES). In this work, Das and Hussain pair cryo-EM with algorithmic RNA structure prediction to present a structure of the Type 2 IRES found in Encephalomyocarditis virus (EMCV). Using medium to low resolution cryo-EM maps, they resolve the overall shape of a critical domain of this Type 2 IRES. They use algorithmic RNA prediction to model this domain onto their maps and attempt to explain previous results using this model.

Strengths:

- (1) This study reveals a previously unknown/unseen binding modality used by IRESes: a direct interaction of the IRES with the initiator tRNA.*
- (2) Use of an IRES-associated factor to assemble and pull down an IRES bound to the small subunit of the ribosome from cellular extracts is innovative.*
- (3) Algorithmic modeling of RNA structure to complement medium to low resolution cryoEM maps, as employed here, can be implemented for other RNA structures.*

We thank Reviewer 2 for positive and encouraging comments on our work, appreciating our ‘innovative’ approach of using IRES-associated factor to assemble and pull down IRES-bound ribosomal complex.

Weaknesses:

- (1) Maps at the resolution presented prevent unambiguous modelling of the EMCV-IRES. This, combined with the lack of any biochemical data, calls into question any inferences made at the level of individual nucleotides, such as the GNRA loop and CAAA loop (Figure 4).*

We understand the concerns raised by the reviewer related to the resolution of the EMCV IRES-48S PIC map. However, we would like to mention that we refrained from commenting on individual nucleotides or molecular interactions in the manuscript. Instead, we discuss about loops, RNA stretches or motifs that could be inferred with more confidence as shown in Manuscript Figure 4. The EMCV IRES can directly interact with the 40S ribosome using its domain H and I (Chamond et al 2014), however, the details this interaction was unknown. We observe that the CAAA loop of domain I apex interacts with 40S ribosome based on the placement of portion of domain I in the cryo-EM map. This is also reflected in the earlier reported SHAPE data (Supplementary figures 2, and 8 in Chamond et al 2014), where a decrease in reactivity is evident in the presence of 40S ribosome. In addition, incubation of EMCV IRES with rabbit reticulocyte lysate (RRL) offered protection to domain I apex regions, which included the CAAA loop (Figure 4b in Maloney and Joseph, 2024).

Furthermore, this decrease in SHAPE reactivity pattern is also evident for FMDV IRES domain 3 apex (like domain I in EMCV IRES) in the presence of 40S ribosome (Lozano et al 2018).

Thus, these studies are consistent with the placement of IRES model in the cryo-EM map.

We aim to improve the resolution of the maps for better clarity and add biochemical experiments to justify the possible interactions.

- (2) The EMCV IRES contains an upstream AUG at position 826, where the PIC can assemble (Pestova et al 1996; PMID 8943341). It is unclear if this start codon was mutated in this study. If it were not mutated, placement of AUG-834 over AUG-826 in the P-site is unexplained.*

We thank the reviewer for bringing up this point, as we missed mentioning this in the manuscript. The EMCV IRES does not require scanning and directly positions the AUG-834 at the P site (Pestova et al 1996). In Pestova et al 1996, the intensity of the toeprint at AUG-834 is much more intense than that of AUG-826. Further, AUG-834 lies in the Kozak context, whereas AUG-826 has a poor Kozak context. Furthermore, the synthesis of the polypeptide requires placement of AUG-834 at the P site. In our cryo-EM map, we observed that the tRNA_i is in a P_{IN} state, which indicates the recognition of the start codon, and we reasoned that it is more likely that AUG-834 is placed at the P site than AUG-826. We will mention this in the revised manuscript, as we had NOT mutated AUG-826.

(3) The claims the authors make about (i) the general overall shape and binding site of the IRES, (ii) its gross interaction with the two ribosomal proteins, (iii) the P-in state of the 48S, (iv) the rearrangement of the ternary complex are all warranted. Their claims about individual nucleotides or smaller stretches of the IRES-without any supporting biochemical data-is not warranted by the data.

We thank the reviewer for warranting major claims, and we wish to make further improvements to support our assessment of small stretches and individual nucleotides.

Reviewer #3 (Public review):

Summary:

Type II IRES, such as those from encephalomyocarditis virus (EMCV) and foot-and-mouth disease virus (FMDV), mediate cap-independent translation initiation by using the full complement of eukaryotic initiation factors (eIFs), except the cap-binding protein eIF4E. The molecular details of how IRES type II interacts with the ribosome and initiation factors to promote recruitment have remained unclear. Das and Hussain used cryo-electron microscopy to determine the structure of a translation initiation complex assembled on the EMCV IRES. The structure reveals a direct interaction between the IRES and the 40S ribosomal subunit, offering mechanistic insight into how type II IRES elements recruit the ribosome.

Strengths:

The structure reveals a direct interaction between the IRES and the 40S ribosomal subunit, offering mechanistic insight into how type II IRES elements recruit the ribosome.

Weaknesses:

While this reviewer acknowledges the technical challenges inherent in determining the structure of such a highly flexible complex, the overall resolution remains insufficient to fully support the authors' conclusions, particularly given that cryo-EM is the sole experimental approach presented in the manuscript.

The study is biologically significant; however, the authors should improve the resolution or include complementary biochemical validation.

We thank Reviewer 3 for acknowledging the technical challenges in this study and finding our study biologically significant. We understand the concerns related to low resolution and the requirement of complementary biochemical validation for our reported observations and interpretations in the manuscript. We are attempting to improve the resolution and complement the interpretations with biochemical experiments.

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